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Balancing privacy and usability: A design science research approach for cookie consent mechanisms

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ABSTRACT

Digital analytics literature has received increasing research attention regarding website cookies owing to the implementation of the General Data Protection Regulation (GDPR) and its role in online privacy decisions and data collection. However, there is still a gap in the literature about users' perspectives and beliefs about the intention to accept (opt-in) to the website cookie consent. Therefore, this study focuses on measuring user behavior toward website cookie consent mechanisms by integrating the Technology Acceptance Model (TAM), Protection Motivation Theory (PMT), and Design Science Research (DSR). Through a survey of 384 participants aware of website cookie consent in Jordan, this study examined the influence of factors such as perceived usefulness, ease of use, and threat and coping appraisals on accepting data collection through website cookies while browsing the Internet. The findings highlight the higher influence of perceived ease of use on driving consent, highlighting the need for consent that is user-friendly while protecting data privacy. By integrating DSR, this study incorporates open innovation principles by proposing innovative design requirements for website cookie consents. In addition, it offers valuable insights for both academics and practitioners by proposing a starting point for further investigation of technology acceptance, online privacy decisions, and data collection privacy regulations.

1. Introduction

With the rapid advancements in digital analytics, businesses are gathering information about users who browse their websites (Núnez-Barriopedro et al., 2023). This information has led to growing attention being paid to online behavioral advertising (OBA), personalized targeting, and user data protection. Therefore, data privacy concerns were triggered by the introduction of the General Data Protection Regulation (GDPR) owing to data collection through website cookies (Lukic et al., 2023). The GDPR established fines for businesses that collect user data without obtaining proper consent from the users. Additionally, the GDPR forces self-determination by imposing transparency requirements for business data collection practices and strengthening individuals' rights to their personal data (Utz et al., 2019). Therefore, businesses took proactive measures on their websites by requesting explicit consent from users before collecting and processing the website data. This website consent is applied using cookie notices (also known as cookie consents or banners), which allow users to consent while showing options, such as accepting all website cookies, accepting specific website cookies, or declining all website cookies. The decision lies solely with the user, and it is the website's responsibility to ensure proper implementation of the user's choice.

The GDPR requires the implementation of an opt-in (i.e., acceptance of participation in data collection) model to govern cookie notices. Therefore, users have the option to dismiss the cookie notice without making any custom choices, permitting essential cookies, or opt-out (decline) all-tracking (Giese and Stabauer, 2022). Some websites posit that particular cookies are essential for their functioning (for example, to deliver the page) or because of legitimate interests (such as product improvement), and therefore cannot be declined by users (Papadogiannakis et al., 2021). Websites heavily depend on advertising and web analytics to customize their offerings. To achieve personalized targeting, web analytics collects data to send personalized ads based on the information gathered by HTTP cookies (Urban et al., 2020). Therefore, website owners depend on increasing the quantity of users who opt-in into cookie notices to gather as much data as possible about user

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behavior (Giese and Stabauer, 2022). Therefore, encouraging website users to opt-in into cookie notices is a significant research topic (Schmidt et al., 2020).

Previous research has explored users' perspectives on cookie consent positions on webpages (Utz et al., 2019), choice architecture of consent (Bauer et al., 2021), language of cookie notices, users' perception of GDPR compliance (Marikyan et al., 2021), perceived control of data collection, users' technical and legal knowledge toward the motivation to reject cookies (Strycharz et al., 2021), and the effectiveness of existing cookie consent mechanisms in protecting user privacy (Degeling et al., 2019). Despite these contributions, a significant gap remains in the literature regarding users' intentions to accept (opt-in) cookie consent, particularly within the framework of users' perceptions of the ease of use and usefulness of cookie consent mechanisms, as examined by the Technology Acceptance Model (TAM). In addition, factors such as threat and coping appraisals, as defined by Protection Motivation Theory (PMT), have not been empirically explored in the context of cookie consent. While existing studies emphasize the importance of data collection and privacy, there is a notable lack of empirical investigation into how these psychological factors shape user engagement with cookie consent mechanisms.

To address this research gap, this study seeks to answer the following research questions:

RQ1. How do the constructs of perceived usefulness and perceived ease of use, as defined by TAM, influence users' intentions to accept (opt-in) cookie consent?

RQ2. How do threat and coping appraisals, as outlined by the PMT, affect user decisions regarding cookie consent?

RQ3. How can cookie consent mechanisms be designed to enhance user friendliness, while ensuring privacy protection?

In alignment with these research questions, the study proposes the following hypotheses:

H1. Perceived usefulness of tracking cookies positively affects users' motivation to accept (opt-in) tracking cookies.

H2. Perceived ease of use in controlling tracking cookie preferences positively affects users' motivation to accept (opt-in) tracking cookies.

H3. Perceived severity of data tracking risks moderates users' motivation to accept (opt-in) tracking cookies.

H4. Perceived susceptibility to data tracking risks moderates users' motivation to accept (opt-in) tracking cookies.

H5. Self-efficacy in managing data collection and processing through cookies moderates users' motivation to accept (opt-in) tracking cookies.

H6. Perceived efficacy of website tracking consent moderates users' motivation to accept (opt-in) tracking cookies.

This study incorporates open innovation principles by proposing innovative design requirements for website cookie consents. Employing Design Science Research (DSR) as the methodological framework, it develops novel cookie consent mechanisms that not only address the ease of use and perceived usefulness of cookie consent, but also consider users' threat and coping appraisals. A Requirements-Focused DSR approach is used to systematically gather requirements and design solutions, ensuring that the resulting artifacts are empirically grounded in user insights. By integrating DSR, this study offers practical contributions through the development of more effective, user-friendly, and privacy-preserving cookie consent mechanisms. It also contributes theoretically by integrating TAM and PMT into the fields of privacy research, information systems, and consent management.

The structure of this paper is as follows: section two presents a background on website cookies and GDPR, section three presents related work and hypothesis development, section four presents the research methodology, followed by the results in section five, discussion in section six, and the conclusion in section seven.

2. Background

2.1. Website cookies and general data protection regulation (GDPR)

HTTP website cookies are snippets of text embedded within browsers upon website visits and are designed to facilitate information sharing. Cookies empower website providers to store data on user devices and enrich the user experience by facilitating specific website functionalities (Miyazaki, 2008). However, the role of cookies extends further because of granting website providers insight into user interactions. These insights are used to develop personalized user profiles, which are the cornerstones of tailoring advertisements to individual preferences (Kulyk et al., 2018). When a user explores a new website, cookies and website tracking IDs are allocated, enabling both first and third parties to re-identify user profiles across the web. These profiles may subsequently be stored on data-management platforms (Elmeleegy et al., 2013), marketed by data brokers, or used by advertisers to participate in ad auctions (Haque et al., 2022), execute ad retargeting, or conduct cross-device tracking (Iordanou et al., 2019; Solomos et al., 2019).

Cookies are categorized according to the European advisory body on data protection and privacy in (Article 29 Working Party, 2012) into two main categories: a) session cookies and persistent cookies and b) first-party and third-party cookies. The use-case scenarios elaborate on these categories as follows.

- User-input cookies: refer to first-party session cookies that track user input during interactions, are often used for online forms and shopping carts, typically relying on a session-ID (a random temporary unique number), and expires when the session ends at the latest.
- The authentication of cookies enables user identification after login (e.g., online banking). These session cookies facilitate repeated authentication to access authorized content such as account details during subsequent site visits.

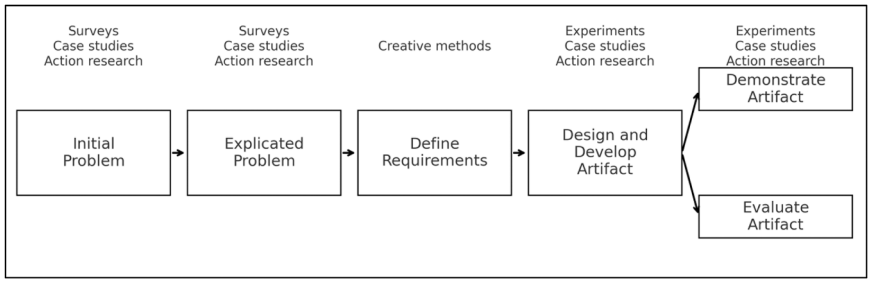


Fig. 1. Overview of design science research.
Adapted and reworked from Johannesson and Perjons (2022).

- Third-party ad cookies are not essential, and by the data protection regulation privacy by design principle, they conflict with the General Data Protection Regulation (GDPR).
- First-party cookies were used to track the audience engagement. Despite not being considered strictly essential, these cookies pose minimal privacy risks when properly utilized.
- User-centric security cookies: Cookies employed for identifying recurrent unsuccessful login endeavors on-site or comparable mechanisms intended to fortify the login system against misuse, with potentially modest efficacy.
- Multimedia player session cookies (often called "flash cookies) retain technical video or audio playback details such as image quality and buffering. Typically, they expire by the end of a session because of their temporary relevance.

Kretschmer et al. (2021) categorized cookies as functional or nonfunctional. Functional cookies utilized within the HTTP protocol of web services address the challenge of transmitting all data required for rendering a website during each request. These cookies primarily serve to maintain configurations across multiple requests such as language preferences for international sites. While some functional cookies, such as preference settings, remain non-personal, others, such as authentication cookies, can identify users. This category is often collectively referred to as "trackers." For example, after a user log into a web service, a random string is sent to the browser for verification. Providers can use these cookies for user behavior analysis and personalized recommendations. Nonfunctional cookies, often employed by third parties, lack functional purposes. These cookies aim to create browsing profiles that can potentially reveal user identities on specific sites. Users can reject these cookies without compromising service usability.

Third-party trackers are primarily used in customized advertising. Because third-party tracking can involve all browser sessions and share data outside the sites you visit, it poses serious privacy issues. This places greater pressure on advertisers and website owners to collect as much personal information as possible. Recently, the importance of data collection and utilization in tailored advertising has increased. It is crucial for online shop operators, especially in e-commerce, to retain specific data in cookies to enhance the site's usability and evaluate various design and procedure improvements. Cookies aid in the data collection process for different marketing initiatives and conversion improvements (Giese and Stabauer, 2022).

Because website cookies remain on a user's computer, they pose a potential threat to users. Therefore, policymakers increasingly regulate (e.g., the European Union's Regulation [GDPR]) that consumers receive visible notice and choice over the collection of their private information online. In particular, the GDPR requires website operators to be transparent about their cookie-tracking practices by displaying cookie notices, and all users should have the choice to either allow or deny the use of cookies to track their online behavior (Bornschein et al., 2020).

Although cookies serve a useful purpose in the digital world, they also raise privacy concerns (Miyazaki, 2008). Cookies track consumers' browsing behavior on company websites by allowing the cookie originator to identify sources of consumer traffic, tracking consumers' clicks on a website (e.g., through Google Analytics), adjusting its offering (e.g., adjusting prices for returning visitors), and advertising in a more targeted manner (Bornschein et al., 2020). After the implementation of GDPR, websites now ask users for consent prior to setting cookies (Utz et al., 2019; Bellentani, 2020). GDPR's impact of GDPR on web data collection and processing is summarized as follows.

- The Right to Be Forgotten: Article 17 requires data controllers to delete personal data upon request, except when collected through legal bases such as consent or anonymization; data used in systems to train a recommender system is technically infeasible to prove afterward whether this data concerned a specific user and remained unaffected by deletion requests.
- Transparency: Article 15 of the GDPR grants individuals the right to request personal data from service providers. Article 13 mandates that data controllers transparently communicate details of data processing, including purpose, duration, and legal basis. In the case of data breaches, Article 33 requires that individuals be notified that the breach threatens their rights and freedom.
- Data Portability: When a data controller processes personal information based on user consent, the user has the right to receive this information from the controller in a format that is both "common and machine readable," as outlined in Article 20 of the GDPR. This allows users more control over their personal information as they can determine the purpose of data collection and services that can access specific data.

2.2. The technology acceptance model (TAM)

The Technology Acceptance Model (TAM) focuses on user attitudes, intentions, and behavior in order to understand the acceptance of a technology or a system (Davis, 1989) by introducing factors such as "perceived usefulness" known as individuals belief that a particular system is useful and it improves their job performance while using the system. Even when potential users recognize the usefulness of a system, they may simultaneously perceive it to be too complex or difficult to use, with the effort required to outweigh the expected performance benefits. This definition introduces the "perceived ease of use" which refers to "the extent to which a person believes that using a particular system will be effortless."

By mapping perceived usefulness and perceived ease of use from Davis (1989) to website cookie consent mechanisms, we can understand how users approach acceptance or rejection of cookie usage on websites. For example, users evaluate whether accepting website cookies improves their browsing experience and protects their data-privacy preferences. In this context, this is aligned with the definition of "perceived usefulness" and can be presented as the degree to which users believe that consenting to cookies will positively impact their online interactions, ensuring a smoother and more tailored web experience. In addition, even when users acknowledge the potential benefits of allowing cookies, they may perceive this action as complex and difficult. They may fear that the effort required to understand and manage cookie settings will outweigh the expected advantages. This dimension corresponds to "perceived ease of use" in the context of cookie consent. It reflects the extent to which a person believes that managing cookie preferences on a particular website will be effortless, aligning with the "ease" - the absence of difficulty or significant effort.

By adopting TAM for cookie consent mechanisms, a better understanding of how users navigate between the potential benefits and perceived complexities of accepting cookies can be achieved, ultimately influencing their decisions.

2.3. The protection motivation theory (PMT)

Protection Motivation Theory (PMT) is a psychological theory that attempts to understand the choices people make regarding protective action in response to perceived threats or risks. The PMT has been developed and used across many areas, including health and safety, and in this study, online protection behavior. The PMT proposes two cognitive processes that motivate individuals to engage in protective behaviors: threat and coping appraisals.

Threat Appraisal: This evaluates perceived threats, which are analyzed in two aspects within this component of the PMT. These were based on the measures of perceived threat, which referred to how bad a certain threat was, and perceived vulnerability, which referred to the likelihood of threat occurrence. In the context of threats, this may refer to how people interpret the severity and probability of viral threats (Lee et al., 2008). Within the context of cookie consent, threat appraisal encompasses the assessment of perceived threat associated with online

data collection as follows:

- Perceived Severity: This corresponds to people's estimates of the threat and the extent of possible harm they anticipate. When it comes to data gathering on the web, people evaluate the probable risks associated with the breach of data or misuse of their information.
- Perceived Susceptibility: This corresponds to the perception of vulnerability to threats. Web data collection involves a probability assessment of when an individual's data is vulnerable and can be violated.

Coping Appraisal: This one's capacity to cope with threats, including perceived response efficacy (belief that preventive measures are effective) and self-efficacy (confidence that one can perform protective actions). Cookie consent and online privacy focus on an individual's ability to cope with perceived threats (Lee et al., 2008). Coping Appraisal is broken down into two factors.

- Perceived Response Efficacy: This measure of an individual's belief that the action intended to reduce the threat is likely to work as intended. This factor focuses on individuals' trust in a set of recommended protective actions, such as changing their cookie preferences or using privacy tools, which can effectively prevent online privacy threats.
- Perceived Self-Efficacy This measure of an individual's belief in their ability to take protective action in the context of online privacy and cookie consent. In addition to users' willingness to trust that they can manage and control their own online data and therefore adjust cookie settings, they use browser extensions to improve privacy.

2.4. Design science research

Design science research (DSR) is an innovative process for creating and designing artifacts to solve real-world problems (Apiola and Sutinen, 2021; Johannesson and Perjons, 2022). DSR is the practice of designing and producing artifacts to meet certain requirements and then testing and validating them to ensure appropriate functionality. An artifact is an object created by a human for practical reasons. For example, the IT and information systems discipline involves a variety of artifacts such as algorithms, programs, systems, and software architectures (Johannesson and Perjons, 2022).

The steps in DSR research are more rigorous at every stage and provide greater depth and precision throughout the research process. This rigorous application helps address all stages of the project, from problem identification to solution evaluation (Apiola and Sutinen, 2021). DSR projects are broken down into phases, such as problem explication, requirement definition, design and development, demonstration, and evaluation (Johannesson and Perjons, 2022). In particular, design science research offers a holistic methodology for real-world problem solutions for sociotechnical systems, including social and technical dimensions, thus requiring a method that unifies both the social and technical dimensions to create feasible solutions. The phases are described in Fig. 1 and summarized as follows:

- Explicate Problem: This phase begins with the identification of a problem within a sociotechnical system, which often involves understanding the interaction between the human (social) and technological components. This step requires a deep understanding of the social practices, workflows, and cultural elements that interact with technological tools and systems.
- Define Requirements: In phase, the problem to be tackled is created as a solution-specified problem in the form of an artifact that elicits the requirements. These requirements satisfy both functional and non-functional requirements. Functional requirements are a set of services or requirements for a given system to deliver, how it should respond to a particular input, and how it should operate under defined circumstances. Nonfunctional requirements do not pertain to specific services provided by the system to its users. Instead, they address broader system qualities such as the speed of response, security, and dependability. These requirements often surpass individual functional requirements that users may be able to work around if they are not fully satisfied. Failure to meet a nonfunctional requirement may render the entire system unusable, such as in aircraft or embedded control systems. Examples of non-functional requirements include product requirements that define software behavior, including performance and reliability requirements (Sommerville, 2015).

- Design and Develop Artifacts: This activity includes the creation of an artifact that resolves the specified problem and satisfies established requirements. This process involves determining functionality and structure (Johannesson and Perjons, 2022).

- Demonstrate artifact: This illustrates the applicability of an artifact within a relevant context in order to demonstrate its problem-solving ability. This could involve presenting case studies, conducting simulations, or employing other practical approaches that highlight the effectiveness of the artifacts. Evaluation Artifact: The primary function of this activity is to assess the extent to which the artifact satisfies the established requirements and to effectively address the practical issues that triggered the research.

Many design science research studies have not considered all five activities within a given framework. Instead, they may focus on one or two activities such as problem-focused design science, which concentrates on problem explication. On the other hand, requirement-focused design science research typically involves projects that prioritize defining requirements. These projects usually begin with an existing problem that is either accepted or slightly elaborated upon. Researchers have conducted thorough investigations, including literature reviews and stakeholder interactions, to determine these requirements. Other studies may focus on requirement- and development-focused design science research, development- and evaluation-focused design science research, or evaluation-focused design science research.

This research is centered on Requirements-Focused Design Science Research, which involves literature on website cookies, as well as interaction with relevant users through a questionnaire.

3. Related work and hypothesis development

A growing body of literature on website cookie consent mechanisms has received increasing attention, mainly since the introduction and implementation of the GDPR (Giese and Stabauer, 2022). The study by Negri-Ribalta et al. (2024) conducted a systematic mapping study to explore the role of requirements engineering in GDPR compliance. The key findings revealed that GDPR is receiving ongoing research interest and consent management research topic is of the most published topic. In addition to the limited interdisciplinary research between computer science and legal scholars, but many studies link software engineering to GDPR by integrating the software development life cycle with GDPR requirements. Multiple studies have reported that the design of cookie consent banners significantly affects user interactions and decision making. For example, a study by Utz et al. (2019) involved over 80,000 website users, revealing that cookie consent placed at the bottom left of the webpage was more likely to capture user interactions. Additionally, binary choices such as "Accept All Cookies" or "Reject All Cookies" lead to higher opt-in rates as binary choices simplify decision-making for users. Pre-selected cookie options, or nudging, further increased the likelihood of users accepting website cookies, while technical language and policy links had minimal impact on opt-in rates. The study Bauer et al. (2021) conducted a field study on a Danish B2B website and confirmed that manipulating the choice architecture—such as enlarging

Table 1
Comparative analysis of previous studies related to cookie consents.

Study	Focus on Cookie Consent	Scope	Contributions	TAM, PMT, and DSR on Website Consent Mechanisms
Utz et al. (2019)	Binary choices (Accept/ Reject)	Pre-selected options (nudging)	Explains on how binary cookie consent choices increase opt-in rates and how pre-selected options (nudging) further influence user consent.	×
Bauer et al. (2021)	Enlarged "accept" button, positive framing	Choice architecture manipulation	Investigates how manipulating choice architecture (e.g., enlarging the "accept" button) can increase consent rates and raises ethical concerns about nudging techniques that may violate GDPR principles.	×
Strycharz et al. (2021)	Impact of technical/legal knowledge on consent	Knowledge impact on consent decisions	Investigates the impact of technical and legal knowledge on users' willingness to reject cookies. Although such knowledge reduced perceived threat severity, it did not significantly enhance users' motivation to reject cookies.	×
Marikyan et al. (2021)	Factors influencing compliance behavior	Individuals' attitudes about GDPR compliance	Focuses on how perceived threat severity, self-efficacy, and response efficacy influence users' attitudes toward GDPR compliance.	×
Martini and Drews (2022)	Ethical concerns around consent	Dark patterns in consent designs	Explores the ethical implications of dark patterns, such as pre-selected consent boxes and unequal button sizes, which undermine users' ability to give informed and voluntary consent under GDPR.	×
Altpeter (2022)	Consent dialogs in mobile apps	Mobile consent dialogs	Analyzes how dark patterns in mobile app consent dialogs often lead to users consenting without fully understanding the data collection practices, highlighting the need for stricter enforcement of GDPR in mobile apps.	×
Singh et al. (2022)	Global preferences for cookie designs	User preference for banner designs	Studies user preferences for cookie consent designs. Findings reveal that simple designs such as sliders and binary choices are favored over more complex designs such as full vendor lists.	×
Porcelli et al. (2023)	Automated responses to consent banners	Personal information management (PIMS)	Introduces an automated consent management system (PIMS) to minimize user interaction with cookie banners, reducing the influence of dark patterns and improving GDPR compliance.	×
Ogut (2023)	Impact of banner placement on young users	Influence of banner placement	Focuses on how the placement of cookie consent banners influences user behavior, showing that centrally placed banners that block content led to higher acceptance rates.	×
Palm and Jensen (2024)	User experience (UX) of cookie consent	Prototype for better consent management	Proposes a user-friendly and GDPR-compliant prototype that simplifies cookie consent interfaces, making rejection as easy as acceptance. Enhances user autonomy and promotes transparency.	×
Vuorinen (2024)	Hiding "decline" options (No Decline Banner)	Banner design with dark patterns	Examines how dark patterns, such as hiding the "decline" option behind additional layers, reduce users' ability to reject cookies. Transparent designs that highlight refusal options led to higher rejection rates.	×
Bouhoula et al. (2024)	Automated large-scale analysis of cookie notice compliance	Use machine learning models and natural language processing to analyze cookie consents with GDPR	Reported high number of GDPR violations such as using dark patterns, and data collection without user consent.	×
Porcelli et al. (2024)	Cookie banners automation and dark patterns	Use Large Language Models (LLMs) to analyze cookie banners	Propose solutions to reduce the role of dark patterns in cookie banners and automate users' interactions based on predefined preferences.	×
Negri-Ribalta et al. (2024)	A systematic mapping study	Requirements engineering in GDPR compliance	Explored requirements engineering and GDPR compliance and reported that there is limited ongoing research that integrates GDPR and software development lifecycle.	×
Biselli et al. (2024)	The impact of personalised cookie banner	Usability and personalised cookie consent banners	Confirms that dark patterns in cookie consent design have a harmful impact on users' decision making. Instead, the study suggests personalised cookie banners since it increases usability rates.	×
Storbjörk and Jordan (2024)	How to improve user decision making in cookie banner design	Users' interaction with cookie consent banners	Propose design guidelines for user-friendly cookie consent banners	×
Inal et al. (2024)	Cookie consent design and dark patterns	Users' interaction with cookie consent banners	Explores visual attention in cookie consent interface.	×
Clarke et al. (2024)	An empirical analysis on tracking third-party cookies	State of third-party cookies	Explores the privacy risks and security vulnerabilities.	×
Franke et al. (2024)	GDPR in the development of open-source software	The impact of GDPR on open-source software development	Examined the cost, complexity and effort of GDPR compliance in open-source software development.	×
Jha et al. (2024)	Privacy policies and consent management platforms	Longitudinal study on consent management systems and GDPR	Examined users' interactions with privacy polices over time.	×
Berens et al. (2024)	Impact of dark patterns on user consent decision	Impact of dark patterns	Investigated the impact of dark patterns on user consent decision. The findings are important for how website design manipulates user consent decisions	×
Buckley (2025)	Analysis of GDPR's effectiveness	Impact of GDPR	Investigated how businesses benefited from GDPR.	×

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Table 1 (*continued*)

Study	Focus on Cookie Consent	Scope	Contributions	TAM, PMT, and DSR on Website Consent Mechanisms
Rauti et al. (2025)	Data collection and third party analytics in healthcare websites	Possible data leak of user sensitive information	Investigated the privacy threats in healthcare websites in Finland.	×
Matthew et al. (2025)	Privacy and dark patterns	Users' behaviour toward dark patterns	Investigated if consumers can protect themselves against privacy dark patterns.	×
Leith (2025)	Cookies Googlesilently store on Android handsets	Google services and play store	Investigated what cookies, user and device identifiers are collected and stored by Google services on Android handsets. And whether there is a user consent for opt-in or out.	×
This Study	Integrating TAM, PMT, and DSR for website consent designs	Requirements-focused design science	Requirements-focused design science research applied to collect requirements to design and develop innovative website cookie consents that address the ease of use, perceived usefulness, considers users' threat and coping appraisals and proposes GDPR-compliant requirements for consent designs.	✓

the "Accept" button and using positive framing led to a 17 % increase in consent rates. This study also raised ethical concerns about whether these nudging techniques align with GDPR principles.

Dark patterns have initiated debates about the ethics of website cookie consent design. Martini and Drews (2022) argued that these manipulative tactics (i.e., dark patterns) exploit users' behavioral tendencies, undermining the GDPR's goal of informed and freely given consent. These dark patterns, which include pre-selected consent boxes, unequal button designs, and hidden information, create barriers for users attempting to reject cookies and impose users toward consenting to extensive data collection. Martini and Drews (2022) called for policy reforms including banning specific dark patterns and enhancing transparency in website cookie consent mechanisms. Altpeter (2022) explored the implementation of website consent in mobile apps and their compliance with the GDPR and the ePrivacy Directive. Analyzing 4388 popular apps from Android and iOS, the study reported that over 90 % of the website consent banners used dark patterns, which manipulated users into giving consent without fully understanding or voluntarily agreeing to participate in data collection. Furthermore, many apps continue to process website tracking data even when users reject consent, reporting a significant gap between legal requirements and real practices. The study emphasizes the need for stricter enforcement of data protection laws and improvements in the design of consent banners to genuinely maintain user privacy rights. Similarly, other studies have examined the design patterns in different contexts. For example, Vuorinen (2024) and Bielova et al. (2024) examined the effects of dark patterns on cookie-consent banners. Their research revealed that dark patterns, such as hiding the "Decline" option behind additional layers (No Decline Banner), reduce the likelihood of cookie rejection, whereas transparent designs that highlight refusal options (Highlighted Decline Banner) or emphasize the risks of tracking (Consequence Banner) lead to significantly higher refusal rates.

The study by Porcelli et al. (2024) reported that most users undergo fatigue due to repetitive cookie interactions, and most websites use dark patterns in cookie consent design to force users to opt-in. Therefore, this study uses Large Language Models (LLMs) to propose a system that automates user interactions with cookie banners based on predefined preferences. This system can reduce the effort required by users to interact with cookie banners and diminish the impact of dark patterns.

The study by Biselli et al. (2024) reported that dark patterns and complicated cookie banners add more challenges to users' decision-making when dealing with cookie banners. Based on users' privacy knowledge, this study emphasizes empirical evidence that personalized cookie banners enhance both usability and privacy protection. Inal et al. (Inal et al., 2024) reported that users do not read cookie consents; instead, they are more likely to select easy options. In addition, the study reported that dark patterns in cookie consent designs influence user decisions. This suggests that UX designers, web developers, and policy makers should design improvements to improve user perception and transparency. The study by Berens et al. (2024) examined the impact of dark patterns on user consent. While websites collect unnecessary data through cookies without user consent, the study highlighted that visual tactics, including dark patterns, impact user decisions and recommend web developers to enhance transparency in website cookies.

The study by Matthew et al. (2025) examined user behavior in dark pattern contexts. Experimental evidence revealed that users struggle to navigate such design choices and illustrated that cookie consent mechanisms, including the opt-out option, can help users counteract dark patterns. The study by Bouhoula et al. (2024) used machine learning models and natural language processing (NLP) to analyze website compliance with GDPR. The study revealed that high violations, such as the cookie reject option, will still collect data, cookie notices with a missing opt-out option, and the widespread use of dark patterns that manipulate user decision-making.

Using the Protection Motivation Theory (PMT) framework, Marikyan et al. (2021) investigated adherence to website cookie consent regulations. Their findings revealed that attitudes toward GDPR compliance were positively influenced by three factors: perceived severity of threats, individuals' belief in their ability to comply, and perceived effectiveness of compliance measures. However, the response cost and threat vulnerability were found to have less of an effect. The same is true, for example, in Strycharz et al. (2021), who examined whether providing users with technical and legal information about data collection affects their willingness to reject website cookies. According to their study, even such knowledge reduced perceived threat severity but did not significantly improve users' tendency to reject website cookies in general, and even slightly increased self-efficacy. There is variance in website cookie consent design preferences worldwide. Singh et al. (2022) studied website cookie consent preferences across 30 countries and showed that users prefer simple and transparent consent services such as sliders or binary choices. They found it difficult to navigate more complex designs, such as full vendor lists with no clear "Reject All" option, limiting users' ability to make informed privacy decisions. Other studies, such as that of Porcelli et al. (2023), have offered solutions to the overuse of traditional cookie consent. For example, the Personal Information Management Service (PIMS) was introduced to automate user responses to consent banners, reduce the prevalence of dark patterns, and enhance GDPR compliance. In addition, Bruhner et al. (2023) proposed Advanced Data Protection Control Plus (ADPC +), a browser integrated solution that would allow the user to set a privacy preference once and apply it ubiquitously across all websites, thereby removing the necessity for continual interaction with the cookie consent banner.

The same issues were raised in other regulatory environments

outside the EU, such as the California Consumer Privacy Act (CCPA). The study by Mazumdar et al. (2023) reported that some CCPA banners used complex designs and 'nudges' to keep users perpetually opted in. This highlights the need for transparency and ease of understanding that manipulative tactics are pervasive. Other studies, such as those by Ogut (2023), Palm and Jensen (2024), and Puhtila et al. (2024), also underscore that the design of cookie consent banners dramatically affects user engagement. Higher acceptance rates were observed when banners blocked content, as users quickly accepted cookies to remove intrusive banner text but (often) did not read the information. This causes issues related to whether such designs deliver true informed consent according to GDPR requirements. Other studies, such as Puhtila et al. (2024), identified issues of dark patterns that undermine user autonomy, and Palm and Jensen (2024) developed a user-friendly consent management prototype adherent to the GDPR. According to Habib et al. (2022), dark patterns often force users to accept data collection by default, and consent banners that allow users to control their choices are more effective at promoting privacy-protective behavior.

In cookie consent banners, studies by Gerber et al. (2023) and Bruhner et al. (2023) analyze nudging techniques, finding that opt-in settings reduced cookie acceptance compared to opt-out defaults, but that nudges such as color highlighting or timers had little impact on final consent decisions. Nevertheless, these studies called for additional work on design strategies that elevate more informed and autonomous decision making while balancing the ethical implications of the use of nudges. Bergesen et al. (2021) and Graßl et al. (2021) have studied deceptive design elements such as pre-checked boxes, requiring multiple clicks to decline cookies, and obvious" accept' buttons that often lead users to make choices regarding their privacy that are not in their best interest. They run contrary to the goals of the GDPR by reducing transparency and limiting users' ability to make informed choices about using their personal data.

Ma and Birrell (2022) studied how the framing of cookie consent banners affects user decisions. Drawing on Prospect Theory, this study examines two main risks associated with cookies: functional risk (e.g., denying cookies could hurt user experience) and privacy risk (e.g., accepting cookies would allow websites to access and sell personal data). The authors conducted an empirical user study with 1557 participants interacting with different cookie-consent banners. The results showed that negative framing, that is, highlighting losses, was more effective in inducing users to discontinue when functional risks were highlighted, and at lowering opt-in rates when privacy risks were prominent. The study also demonstrated that cookie opt-out rates changed by a factor of three when combining slant (i.e., positive or negative) and framing (i.e., gain or loss). These findings suggest that psychological techniques in cookie consent should not be used in cookie consent interfaces and that such use should be regulated carefully to ensure informed and voluntary user consent.

Other researchers have discussed various aspects of cookie consent mechanisms as follows:

- The study by Storbjörk and Jordan (2024) focused on the design of cookie consent banners by proposing guidelines for designing more transparent and user-friendly banners.
- The study by Clarke et al. (2024) conducted an empirical analysis to track third-party cookies. The findings reported privacy risks and security vulnerabilities for websites that use third-party cookies, despite privacy regulations.
- The study by Franke et al., (2024) investigated an exploratory mixed-methods study on GDPR compliance using an open-source software. The findings revealed that GDPR compliance in open-source software increased the development effort. Therefore, open-source software developers have a negative experience with GDPR and consent mechanisms owing to compliance complexity and cost.
- The study by Jha et al., (2024) investigated a longitudinal study on GDPR and user consent behavior. The findings revealed that GDPR has increased the adoption of content management systems; 90 % of users opt-in because of frustration, and iOS users have higher opt-in rates than Android because of Apple regulations.
- The study by Buckley (2025) reported a comprehensive analysis of GDPR's effectiveness of the GDPR for consumer privacy rights and data protection. The results show that businesses have benefited from GDPR's implementation of GDPR by improving data management and security. In addition, this study proposes key performance indicators (KPI) to assess the effectiveness of GDPR.
- The study by Rauti et al. (2025) investigated data collection on healthcare websites in Finland. The findings revealed a data leak of user-sensitive information sent to third-party tools through Google Analytics, where users were not informed about data collection and raised privacy concerns because of sharing data with third parties and missing consent.
- The study by Leith (2025) investigated the data stored by Google Play Services, the Google Play store, and other pre-installed Google apps on Android handsets. The study found that Google services collect and store various cookies, such as advertising analytics and A/B testing cookies, the Google Android ID as a user and device identifier, and more without user consent. The study highlighted that this raises concerns about data collection violations against the GDPR, and suggested stricter enforcement mechanisms.

Based on the literature and related work, this study presents a new approach that integrates the Technology Acceptance Model (TAM), Protection Motivation Theory (PMT), and Design Science Research (DSR) to develop a holistic perspective on user acceptance of cookie consent mechanisms in the GDPR framework (see Table 1 for comparison). Although previous studies have primarily investigated the design characteristics of consent banners, such as position, color, and the effect of dark patterns on user behavior (Utz et al., 2019; Bauer et al., 2021; Vuorinen, 2024; Bielova et al., 2024), this study contributes to the body of knowledge by identifying the psychological and motivational influences of users' interactions with cookie consent on websites. While previous studies have highlighted the ethical concerns and manipulative nature of dark patterns in consent (Martini and Drews, 2022; Puhtila et al., 2024) and the search for malfunctions in cookie consent designs (Habib et al., 2022; Graßl et al., 2021), this study contributes to a solution-oriented perspective by integrating DSR to propose design requirements for new website cookie consent.

This study contributes additional value by bridging the gap between existing research emphasizing design manipulation (e.g., Singh et al., 2022; Gerber et al., 2023) and a deeper understanding of technology acceptance and user motivation. By incorporating the TAM and PMT, this study highlights the influence of perceived threat severity and coping mechanisms, demonstrating the design and testing of privacy-enhancing consent mechanisms through DSR.

3.1. *Predicting perceived technology using TAM*

In the subsequent sections, this study examines two key aspects: the Perceived Usefulness of tracking cookies, and the Perceived Ease of Use in controlling tracking cookie preferences.

3.1.1. *Perceived usefulness of tracking cookie*

The impact of cookie consent on user behavior and its effectiveness in delivering adequate information for informed decisions remains uncertain. Kulyk et al. (2018) conducted an exploratory user study to examine how users perceive cookies upon encountering their consent, responses, and various factors that influence their choice to continue using or leave the website. Users often dismiss or disregard cookie consent, fail to grasp its meaning and benefits, hold suspicions owing to perceived opacity in data usage, lack awareness of privacy implications,

or misunderstand the significance of cookie collection. User understanding can be improved by increasing the level of website transparency regarding cookie data usage. Nevertheless, simplified privacy policies have not been widely or effectively applied in many contexts. Comprehension issues with cookie notices, data usage transparency concerns, and unawareness of how cookies are used may lead users to avoid cookie notices because of the perceived lack of perceived benefit in engaging with website cookies (Kulyk et al., 2018). The limited deployment of simplified privacy policies confirms that improving users' understanding and transparency remains very difficult. Bauer et al. (2021) studied the effect of choice architecture on user decisions regarding the sharing of private online data. For instance, they examined how different design strategies shape the disclosure of personal information on the internet. The study argues that choice architecture in online data privacy is important and should encourage user-informed decision making through attractive and easy-to-use acceptance options and positive representations of website cookies benefits (e.g., cookies will enhance website experience). In their study, Giese and Stabauer (2022) investigated the elements that influence user behavior when presented with cookie consent. The findings reveal that users choose to accept cookies (opt in) to close the banner. In addition, most users had unfavorable opinions regarding cookie notices, particularly when they were under time pressure, which consequently affected their data-collection decisions. Bellentani (2020) also suggests that users are generally prone to sharing personal data and are willing to allow access to information when appropriately provided with an incentive such as a benefit.

Accordingly, based on the literature discussed above, we suggest that the Technology Acceptance Model (TAM) is a useful framework for explaining users' decisions concerning online privacy and cookie consent. The TAM posits that user intentions to adopt technology are driven mainly by Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Perceived Usefulness (PU is the users' belief that a technology will improve their performance, and it is relevant to online privacy and cookies because it determines how users will decide to consent to cookie use. With online privacy becoming increasingly important, websites often ask users for permission to use cookies to improve their functionality and provide personalized content. By applying the TAM model to users' behavior on website cookie consent, perceived usefulness will explain users' motivation to accept and provide cookie consent. Therefore, the TAM framework can bridge the gap between users' perceptions of technology utility and their decision making regarding online privacy and cookies. It provides a structured approach for assessing whether users' recognition of the benefits of cookies directly affects their willingness to accept them, thus contributing to a comprehensive understanding of the dynamics of online privacy choices. Thus, we propose the following hypotheses:

H1. Perceived usefulness of tracking cookies positively affects users' motivation to accept (opt-in) tracking cookies.

3.1.2. Perceived ease of use in controlling tracking cookie preferences

Cookie notices enable consumers to choose which information to share with a website (e.g., consenting to or declining the usage of specific cookies). Choice is not mandatory, and websites have considerable implementation of their cookie consent notices, leading to a large variety of manifestations in websites' cookie notices (e.g., choice vs. no choice) (van Bavel and Rodríguez-Priego, 2016).

Cookie consent notices emerged in the wake of the European Union's Directive 2009/136/EC in The Council of the European Union (2009) and were classified by Kretschmer et al. (2021) into the three most common types as follows:

- Cookie walls (Type 1): website users are typically informed about cookie usage and given the option to only click "accept" to provide consent. Owing to the unequal choices presented, websites that implement cookie walls cannot establish data processing methods based on explicit consent. Such a cookie consent serves to notify users about the tracking of their browsing behavior, yet they often lack an option for users to disagree with this data collection. Users are typically presented with an "OK" button to signify an agreement.
- Binary banner (Type 2): customers have control over data gathering (referred to as "Choice-Cookies"), which they may reject or accept. Article 7 of the GDPR emphasizes that consent must be clear, and that it needs to be obvious to the user which cookies are necessary and cannot be rejected.
- Multiple-choice banners (Type 3): These provide users with more granular control of the type of cookies that they can reject.

Utz et al. (2019) investigated the effect of cookie notice position, number of consent options, and language on visitor behavior. The findings revealed that cookie placement affected user choices, and binary choice cookie notices had much higher consent rates. In addition, Bauer et al. (2021) reported that changes in the design, such as changing the color, can change the ease of use of the cookie consent and increase the opt-in rate by 85 %. The work of Singh et al. (2022) offers findings regarding users' behavior with cookie consent based on real data and an empirical focus on notice content and design. The results revealed that users prefer limited information and short cookie notices, and that users who experienced only an "Accept" button in the notices with the lowest consent rate may perhaps find deciding on default options. The study also suggests that website cookie consent should be clear, concise, nonintrusive, and basically part of the website content and not delaying or blocking the website.

This corresponds with Article 12 of the GDPR, where information on privacy policies must be communicated in a concise, transparent, consistent, understandable, and easy-to-access manner using a clear and simple language. According to Degeling et al. (2019), cookie consent mechanisms pose the risk of not being an effective way of providing real user control to make an informed choice about their personal data. This study highlighted that these mechanisms may or may not always clearly and coherently inform users of the consequences of consenting to cookies. Therefore, there is a need to develop cookie banners that enhance transparency and user-friendly notifications, to ensure that users can easily make meaningful and informed decisions regarding data privacy.

Bornschein et al. (2020) reported that consumers' perceived power and risk significantly influence their responses to cookie notices. For example, when users find that they have greater control over their personal data, they tend to respond more favorably to cookie notices. Moreover, this study revealed that when users perceive a higher level of risk associated with their digital information, their willingness to engage in cookie notices and consent mechanisms is affected. Interestingly, perceived power moderated this relationship, indicating that a stronger sense of control could mitigate the negative impact of perceived risks.

Kretschmer et al. (2021) posited that privacy policies are typically detailed documents that outline how websites collect, use, and share data. The current landscape of cookie banners and privacy policies may not align with user preferences for privacy-preserving choices. Their complexity can present a challenge for users seeking easy-to-understand information to make informed decisions regarding data privacy such as accepting, customizing, and rejecting cookie consent. Users generally prefer low-effort options that allow them to maintain privacy while navigating websites seamlessly.

When designing cookie notices, Giese and Stabauer (2022) posited that website owners should consider showing their users a complicated cookie notice that will annoy them further. Therefore, this can contribute to the phenomenon of choice fatigue due to the complexity of available choices and the level of cognitive effort required for decision making. This influence might prompt individuals with weaker preferences to face greater uncertainty when making decisions (Bauer et al., 2021). Consequently, users tend to overlook these notices because of

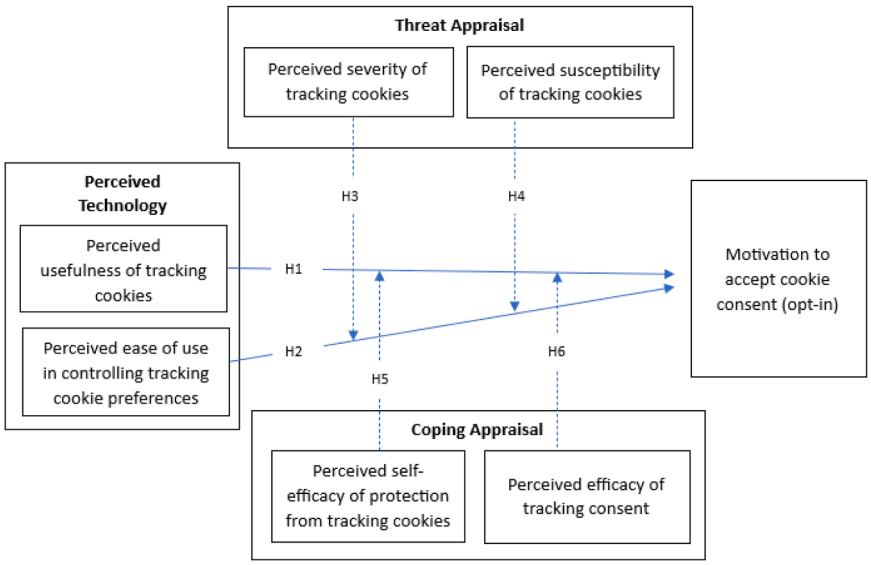


Fig. 2. Research model.

their perceived lack of usability or because they make decisions without a comprehensive understanding of the implications, as observed in the study by Kulyk et al. (2018). Improving the ease of use of cookie banners and privacy policies can provide users with more accessible choices and enhance their control over online privacy.

The extent to which a website enhances the visibility of its information will impact its capacity to fulfill the needs of online customers (Mukherjee and Nath, 2003). Transparency involves making everything visible, signifying an inclination towards open communication to establish trustworthiness (Kim et al., 2014). Multiple scholars have examined transparency in relation to a company's willingness to share information, which can be conceived of as a form of openness within businesses (Parris et al., 2016). Similarly, the perceived ease of use in controlling tracking cookie preferences can be seen as a manifestation of this overarching notion of transparency. Just as organizations strive to provide easily accessible information to foster trust, the design of interfaces and mechanisms for controlling tracking cookies can significantly impact users' perceptions of openness and sense of control over their personal data. Ensuring that individuals find it effortless to manage their tracking preferences reinforces their sense of empowerment, much like transparency reinforces the sense of trust between organizations and their stakeholders. Just as transparency is concerned with making information visible and accessible, perceived ease of use in cookie preference management focuses on making control over data accessible and intuitive, effectively bridging the gap between users and their digital interactions.

Considering the concept of perceived ease of use from the TAM, it becomes apparent that users' assessments of how straightforward and uncomplicated the process of managing their cookie preferences is will influence their decisions. Cookies offer consumers the choice to disclose certain information to websites, allowing them to accept or reject specific cookies. While choice is not obligatory, firms exhibit significant variability in implementing cookie consent notices, resulting in diverse

Table 2
Items of the perceived usefulness of website cookie consents.

Cookie Consent Question on Perceived Usefulness	Corresponding Perceived Usefulness Item from Davis (1989)	Rationale
Information on website cookie consent facilitates decision-making regarding website cookie data collection.	"Using [the system] improves my job performance."	This focuses on the user's perception of how the system helps decision-making, which aligns with improving performance.
I believe that website cookie consents will enhance my overall online experience and personalization.	"Using [the system] increases my productivity."	Enhancing online experience and personalization can be mapped to productivity in the context of digital user experience.
I find that website cookie consent provides me with useful features to improve website functionality and user experience.	"Using [the system] enhances my effectiveness."	Cookie consent functionality improves website experience, similar to effectiveness in system use.
I find that website cookie consents useful to protect my privacy online.	"I find [the system] to be useful in my job."	Privacy protection is seen as a general benefit, similar to how usefulness of a system is perceived in job performance.

Table 3
Items of the perceived ease of use of website cookie consents.

Cookie Consent Question on Perceived Ease of Use	Corresponding Perceived Ease of Use Item from Davis (1989)	Rationale
The process of setting and controlling my cookie tracking preferences is straightforward and user friendly.	"Learning to operate [the system] is easy for me."	This focuses on how easy it is to understand and control preferences, similar to ease of learning to operate the system.
The interactions with website cookie consents are understandable and clear.	"I find [the system] easy to understand."	Directly aligns with how clear and understandable the interactions are.
I find website cookie consent easy to use.	"I find [the system] easy to use."	This is an exact match with the ease of use statement from the original scale.
I often become confused when I use website cookie consents.	"I find it cumbersome to use [the system]."	This corresponds to confusion or complexity, as indicated by cumbersomeness.
Interacting with website cookie consents requires a lot of my mental effort.	"My interaction with [the system] is clear and understandable." / "I find it cumbersome to use [the system]."	Requires mental effort maps to the idea that the system may be difficult or unclear to use.

Table 4
Items of the severity of website cookie consents.

Cookie Consent Question on Perceived Severity	Corresponding Perceived Severity Item from Boerman et al. (2021)	Rationale
Having companies collect my online behavior through website cookies is a problem for me.	"Having companies collect my online behavior is a problem for me."	Focuses on data collection through online tracking.
Having companies use my online behavior through website cookies to show me advertisements is a problem for me.	"Having companies use my online behavior to show me advertisements is a problem for me."	Focuses on the concern of using online behavioral data for targeted advertising.
Having companies share my online behavior with other companies is a problem for me.	"Having companies share my online behavior with other companies is a problem for me."	Directly addresses concerns about data sharing between companies.

Table 5
Items of the perceived susceptibility of website cookie consents.

Cookie Consent Question on Perceived Susceptibility	Corresponding Perceived Susceptibility Item from Boerman et al. (2021)	Rationale
I believe that companies collect information about my online behavior through website cookie consents.	"I believe that companies collect information about my online behavior."	Focuses on concerns of data collection through online tracking.
I believe that companies use information about my online behavior to show me advertisements.	"I believe that companies use information about my online behavior to show me advertisements."	Focuses on the concern that companies utilize behavioral data for targeted advertising.
I believe that companies share information about my online behavior with other companies.	"I believe that companies share information about my online behavior with other companies."	Focuses on data sharing concerns between companies, a key aspect of online privacy threats.

forms of cookie notices such as choice-based versus no-choice options.

Given that cookie consent notices aim to empower users to control their online privacy, it is reasonable to infer that users' perceptions of the ease with which they can exercise control play a significant role. The concept of perceived ease of use from the TAM can be linked to user experience and interaction with cookie consent notices. When users perceive these notices as easy to comprehend, navigate, and utilize, they are more likely to engage in informed decision making and exercise control over their online privacy preferences. Thus, we propose the following hypotheses:

H2. Perceived ease of use in controlling tracking cookie preferences positively affects users' motivation to accept (opt-in) tracking cookies.

3.2. Predicting privacy protective behavior (opt-in motivation) using PMT

Several studies, such as Strycharz et al. (2021), have introduced various elements including perceived security, perceived severity, and perceived susceptibility. These elements collectively shape users' attitudes towards data collection practices, particularly in the context of the General Data Protection Regulation (GDPR). Users' perceptions of their privacy are safeguarded, security and transparency are maintained, and the severity of and susceptibility to potential risks are factors in their decisions to accept or reject tracking cookies.

Protection Motivation Theory (PMT) has emerged as a crucial lens for understanding individuals' responses to privacy and security concerns in the digital landscape. Drawing insights from multiple studies (Hanus and Wu, 2016; Tsai et al., 2016; Zhang et al., 2020; Strycharz et al., 2021; Balaban and Mustățea, 2021), PMT explains users' behaviors, perceptions, and decision-making in various digital contexts. GDPR and online customer trust, as explored by Zhang et al. (2020), delve into the details of perceived threats and efficacy within a regulatory framework. The alignment between PMT and these concepts becomes apparent as users' perceptions of threats regarding their personal data inform their trust-building behaviors and responses to data-protection regulations. Tsai et al. (2016) delved into online safety behaviors from a PMT perspective, illustrating how threat appraisal, self-efficacy, and coping mechanisms shape individuals' decisions to

Table 6
Items of the self-efficacy of website cookie consents.

Cookie Consent Question on Self-efficacy	Corresponding Self-efficacy Item from Boerman et al. (2021)	Rationale
I am able to protect my personal information, such as my browsing behavior while using the internet and interacting with websites that use cookies.	"I am able to protect my personal information, such as my browsing behavior, on the Internet."	Focuses on the ability to protect personal information while online.
I feel confident that I can secure my privacy when it comes to websites collecting my personal information and browsing behavior through cookies.	"I feel confident that I can secure my privacy on the Internet."	Focuses on the individual's confidence in protecting their privacy from websites collecting personal data.
I can ensure that companies cannot collect my personal information and behavior on the Internet.	"I can ensure that companies cannot collect my personal information and behavior on the Internet."	Focuses on the user's ability to prevent data collection.
I have the knowledge to take necessary security measures to protect my personal information and browsing behavior when interacting with websites and their use of cookies on the internet.	"I feel confident that I can secure my privacy on the Internet."	Focuses on the knowledge and ability to take action to protect personal information, extending the idea of self-efficacy.

Table 7
Items of the response efficacy of website cookie consents.

Cookie Consent Question on Response Efficacy	Corresponding Response Efficacy Item from Strycharz et al. (2021), Tsai et al. (2016)	Rationale
I believe that cookie consents would be useful to protect oneself from data collection online.	"I believe privacy protection tools are useful in protecting my personal data" (Strycharz et al., 2021).	Focuses on the added value of a tool (cookie consents or privacy tools) in protecting personal data.
I believe that cookie consents would increase my performance in protecting myself from data collection about your behavior while you browse the internet.	"Privacy protection tools increase my performance in securing my personal data online" (Tsai et al., 2016).	Focuses on the enhancement of an individual's performance in securing their data through protective measures.
I believe that declining to accept cookies through cookie consents is an effective way to protect oneself from data collection online.	"Declining cookies is an effective strategy for protecting personal data online" (Strycharz et al., 2021).	Focuses on the belief in the effectiveness of refusing cookies as a protective action against data collection.

Table 8
Items of the cookie opt-in motivation of website cookie consents.

Cookie Consent Question on Cookie opt-in motivation	Corresponding to the Motivation for Accepting Cookie Consent Item (Strycharz et al., 2021)	Rationale
I intend to click on the accept button (opt-in) in cookie consent.	"I intend to accept cookies on websites that request my consent."	Focuses on the intention to accept cookies and allow data collection.
I see value in allowing websites to collect my online behavior data through cookie consent.	"I believe that accepting cookies on websites has value or benefits."	Focuses on the user's perception of the value or benefit of accepting cookie consent for data collection.
I am likely to accept (opt-in) cookie consent in the near future.	"I am likely to accept cookies on websites that request my consent in the near future."	Focuses on the likelihood of future behavior to accept cookie consent.

adopt protective behaviors. This study underscores the applicability of the PMT in understanding user behavior related to online safety measures, including password management and cautious online behavior.

Extending the PMT to cover desktop security behaviors, Hanus and Wu (2016) explored how users' security awareness affects their actions aligned with the principles of the PMT. Users' perceptions of potential threats to their digital environments and their self-efficacy in mitigating these threats directly influence their security behavior. The literature highlights the adaptability of the Protection Motivation Theory in capturing users' responses to privacy and security challenges in various digital contexts. The integration of PMT enhances our understanding of how users perceive threats, evaluate their coping mechanisms, and ultimately make decisions regarding their online behavior.

3.2.1. *Threat appraisal*

Several studies, such as Moore et al. (2015) and Boerman et al. (2017), report that users are concerned about their online privacy and believe that the customization of advertisements using their personal data is invasive and creative, causing consumer discomfort and violating social norms. In the context of tracking cookies, Strycharz et al. (2021) discussed the threat appraisal concept of PMT (i.e., perceived susceptibility and perceived severity) as a factor of online privacy protection. The research study by Xu (2012) defines threat appraisal as the "concerns about possible loss of privacy as a result of information disclosure.

Threat appraisal is a two-dimensional construct, reflecting the assessment of threat vulnerability and severity. Perceived severity can be defined as a user's judgement of the seriousness of a threat (Rogers, 1975). PMT posits that individuals are more motivated to engage in protective behaviors when they perceive a severe threat. In the context of privacy and security, such as rejecting tracking cookies, if individuals perceive the potential consequences (e.g., loss of privacy or data breaches) to be severe, they are more inclined to take action to protect themselves from potential harm. This perception of severity serves as a driving force for individuals to adopt behaviors aimed at mitigating identified threats. Perceived susceptibility is the extent to which an individual feels that a threat is likely to occur (Rogers, 1975). In this case, potential privacy breaches can occur through the cookie tracking. Individuals determine their vulnerability to the consequences of data collection when granting or denying a cookie consent. If users perceive a high susceptibility to having their online behavior tracked and personal information harvested through cookies, they are more likely to be motivated to make informed decisions regarding cookie consent. This perception of susceptibility influences users' willingness to engage in the cookie consent process and their acceptance or rejection of tracking mechanisms based on their understanding of potential risks.

Hence, in alignment with Protection Motivation Theory (PMT), it is reasonable to assume that the greater the perceived severity and

Table 9
Age distribution for respondents (N = 384).

Age	Frequency	Percent	Valid Percent	Cumulative Percent
20 years or less	116	30.2	30.2	30.2
21–30 years	177	46.1	46.1	76.3
31–40 years	61	15.9	15.9	92.2
41–50 years	25	6.5	6.5	98.7
51 years or above	5	1.3	1.3	100.0
Total	384	100.0	100.0	100.0

Table 10
Level of education for respondents (N = 384).

Level of education	Frequency	Percent	Valid Percent	Cumulative Percent
Bachelor's degree	263	68.5	68.5	68.5
Diploma	15	3.9	3.9	72.4
Postgraduate degree	91	23.7	23.7	96.1
Secondary school or less	15	3.9	3.9	100.0
Total	384	100.0	100.0	100.0

Table 11
Descriptive analysis for items (N = 384).

Construct	Variables	Minimum	Maximum	Mean	Std. Deviation
Perceived Usefulness	PU1	1	7	4.71	1.490
	PU2	2	7	5.03	1.446
	PU3	2	7	4.78	1.415
	PU4	1	7	4.38	1.730
Perceived Ease of Use	PEOU1	1	7	4.70	1.381
	PEOU2	1	7	4.74	1.547
	PEOU3	1	7	4.78	1.415
	PEOU4	1	7	4.25	1.677
	PEOU5	1	7	3.67	1.639
Perceived Severity	PS1	1	7	4.42	1.546
	PS2	1	7	4.01	1.656
	PS3	1	7	5.14	1.617
Perceived Susceptibility	PSUSP1	1	7	5.45	1.265
	PSUSP2	1	7	5.79	1.188
	PSUSP3	1	7	5.14	1.471
Self-efficacy	SE1	1	7	3.96	1.356
	SE2	1	7	3.84	1.519
	SE3	1	7	3.17	1.477
	SE4	1	7	4.30	1.527
Response Efficacy	RE1	1	7	4.45	1.556
	RE2	1	7	4.37	1.408
	RE3	1	7	4.46	1.631
Cookie Opt-In Motivation	OPTIN1	1	7	4.75	1.579
	OPTIN2	1	7	4.24	1.436
	OPTIN3	1	7	4.70	1.400

Table 12
Descriptive analysis for variables (N = 384).

Construct	Minimum	Maximum	Mean	Std. Deviation
Perceived Usefulness (PU)	2.00	7.00	4.7237	1.11401
Perceived Ease of Use (PEOU)	2.25	7.00	4.4704	1.01068
Perceived Severity (PS)	1.00	7.00	4.5263	1.38239
Perceived Susceptibility (PSUSP)	2.00	7.00	5.4605	1.04634
Self-efficacy (SE)	1.00	6.75	3.8191	1.19019
Response Efficacy (RE)	1.00	7.00	4.5614	1.25628
Cookie Opt-In Motivation (OPTIN)	1.00	7.00	4.5614	1.25628

Table 13
Reliability.

Construct	Number of items	Cronbach's Alpha
PU	4	.735
PEOU	5	.884
PS	3	.848
PSUSP	3	.739
SE	4	.859
RE	3	.850
OPTIN	3	.915

susceptibility of website cookies, the higher the likelihood that individuals will take measures to safeguard their privacy. Thus, we propose the following hypotheses:

H3. Perceived severity of data tracking risks moderates users' motivation to accept (opt-in to) tracking cookies.

H4. Perceived susceptibility to data-tracking risks moderates users' motivation to accept (opt-in to) tracking cookies.

3.2.2. Coping appraisal

Coping appraisal, a pivotal aspect of PMT, focuses on evaluating users' beliefs about their capacity to protect themselves, known as perceived self-efficacy, and the effectiveness of the protective measures taken, known as response efficacy. Perceived self-efficacy refers to an individual's confidence in their personal capability to carry out suggested protective behaviors (Rogers, 1975). In other words, it addresses whether an individual believes that they can successfully execute the actions needed to mitigate the threat.

Several academic studies have discussed the relationships between self-efficacy, digital literacy, and digital privacy. Park (2013) recognized that digital literacy, which encompasses the ability to access, evaluate, and communicate information using digital tools, goes beyond technological proficiency. It extends the understanding of the implications of sharing personal data and making informed decisions regarding privacy settings and online behavior. Individuals with higher levels of digital literacy are more likely to exhibit proactive privacy behaviors such as adjusting privacy settings, being cautious about sharing personal information, and comprehending the potential risks associated with online activities.

Boerman et al. (2021) posited that users with less digital experience are more prone to privacy risk. In addition, research has reported that users may not be able to protect themselves, mainly because they do not have knowledge of their personal data. Strycharz et al. (2021) emphasized the significance of technical and legal knowledge as a means of empowering individuals to reject tracking cookies. Strycharz et al. (2021) and Boerman et al. (2021) reported that individuals with a higher level of digital literacy are better equipped to comprehend the details of tracking cookies, including their purposes and potential risks, and are empowered to take protective action. Digital literacy empowers users to make informed decisions regarding online privacy. By being well-versed in the technical mechanisms underlying tracking cookies and their legal rights, individuals can evaluate the potential threats posed by these cookies to their personal information. Furthermore, digital literacy enables users to assess the transparency of data-collection practices. With the ability to interpret privacy policies, terms of use, and the implications of granting consent, digitally literate individuals can critically evaluate whether tracking cookies aligns with their privacy preferences. This empowerment leads to a higher likelihood of rejecting tracking cookies, because digitally literate users are more conscious of their online privacy and the potential consequences of data collection.

Table 14
Model summary for direct effect hypotheses.

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.484[a]	.234	.224	1.10682

[a] Predictors: (Constant), PEOU, PU.

Table 15
Coefficients of direct effect hypotheses.

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.556	.466		3.338	.001
	PU	.172	.091	.153	1.895	.060
	PEOU	.490	.100	.394	4.893	.000

a. Dependent Variable: OPTIN.

Based on the literature, it is evident that self-efficacy plays a pivotal role in shaping users' attitudes and actions towards tracking cookies. As individuals become more skilled at navigating the digital landscape, understanding technical details, and comprehending legal implications, they gain the capacity to confidently reject tracking cookies and assert their privacy rights (Büchi et al., 2017). Strycharz et al. (2021) illustrate the significance of informed decision making driven by technical and legal knowledge. Thus, we propose the following hypotheses:

H5. Self-efficacy in managing data collection and processing through cookies moderates users' motivation to accept (opt-in) tracking cookies.

Within the context of PMT, response efficacy pertains to an individual's perception of the effectiveness of recommended protective measures. This involves assessing whether the suggested actions reduce or eliminate perceived threats (Rogers, 1975). According to Protection Motivation Theory (PMT), individuals must have confidence in the effectiveness of their response to be motivated to engage in protective behaviors (Rogers, 1975). For instance, in the context of safeguarding against computer viruses, research conducted by Lee et al. (2008) revealed that beliefs about the effectiveness of virus protection measures increased the likelihood of adopting such measures to reduce or eliminate the adverse consequences of an Internet virus. In the context of this study, the term 'response efficacy' refers to the perception of the effectiveness of recommended protective actions or strategies in reducing threats to online security. In simpler terms, it explores whether individuals have confidence in the ability of the measures they can take to effectively protect themselves and their online activities from potential risks and harm.

The research by Balaban and Mustățea (2021) explores the link between response efficacy and privacy protective behaviors among mobile users. Individuals who perceive that the privacy-enhancing measures they adopt are effective (high response efficacy) are more likely to engage in privacy-protective behaviors such as adjusting privacy settings, using encryption, or being selective about shared information.

Marikyan et al. (2021) explored the relationship between response efficacy and compliance with the GDPR. Individuals who perceive that the recommended actions and measures, such as consent management or data deletion requests, are effective (high response efficacy) are more likely to comply with GDPR requirements. When individuals are confident about the effectiveness of GDPR-related measures, they may experience lower levels of concern regarding potential data breaches or

Table 16
Summary model for the moderating effect hypothesis (PS).

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.464[a]	.216	.210	1.11634	.216	41.233	1	150	.000
2	.559[b]	.312	.303	1.04880	.097	20.939	1	149	.000

[a] Predictors: (Constant), PEOU.
[b] Predictors: (Constant), PEOU, PS.

Table 17
Coefficients for the moderating effect hypothesis (PS).

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.981	.412		4.810	.000
	PEOU	.577	.090	.464	6.421	.000
2	(Constant)	3.694	.538		6.861	.000
	PEOU	.488	.087	.392	5.625	.000
	PS	− .290	.063	− .319	− 4.576	.000

a. Dependent Variable: OPTIN.

privacy violations. Response efficacy can play a significant role in shaping individuals' trust in organizations that handle their data. Organizations that effectively communicate and implement GDPR-compliant measures can enhance customer trust by reinforcing their perceptions of response efficacy.

In the context of cookie consent, response efficacy pertains to the perception of the effectiveness of the recommended cookie management actions or strategies for mitigating online privacy risks. Essentially, it examines whether individuals have confidence in the measures they can take, such as accepting only essential cookies or adjusting privacy settings, to safeguard their personal data and online activities from potential privacy breaches and harm caused by extensive data tracking.

Therefore, individuals' willingness to actively engage in cookie consent, customize their preferences, and make informed choices about data collection and tracking is tied to their perceptions of response efficacy. In simpler terms, do individuals believe that the actions they take regarding cookie consent genuinely protect their privacy and control the data being collected about their online behavior? Understanding the role of response efficacy in managing cookie consent is pivotal for organizations and policymakers. This underscores the importance of clear and transparent communication regarding the consequences of various cookie consent options, empowering individuals to make informed decisions that align with their privacy preferences. Essentially, response efficacy serves as a critical element in fostering a privacy-conscious digital environment in which individuals are confident that their actions regarding cookies genuinely safeguard their online privacy and data.

H6. Perceived efficacy of website tracking consent moderates users' motivation to accept (opt-in) tracking cookies.

Based on the previous theoretical discussion, this study proposed the research model shown in Fig. 2. This model consists of constructs identified from the literature review as follows: the perceived technology which includes perceived usefulness and perceived ease-of-use as independent variables, threat appraisal which includes perceived severity and susceptibility and coping appraisal which includes self-efficacy and perceived efficacy as moderator variables, and cookie acceptance (opt-in) motivation as the dependent variable.

Table 19
Coefficients for the moderating effect hypothesis (PSUSP).

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.981	.412		4.810	.000
	PEOU	.577	.090	.464	6.421	.000
2	(Constant)	.812	.582		1.394	.166
	PEOU	.548	.089	.441	6.192	.000
	PSUSP	.238	.086	.198	2.781	.006

a. Dependent Variable: OPTIN.

4. Research methodology

4.1. Research design and data collection

This study employed quantitative research methodology using an online survey questionnaire to collect data and test the hypotheses presented in the conceptual model illustrated in Fig. 2. The survey included closed-ended questions and employed a 7-point Likert scale (ranging from 1 = strongly disagree to 7 = strongly agree) to measure the constructs. The questionnaire was developed based on validated scales from previous studies on TAM and PMT, such as Tsai et al. (2016), Strycharz et al. (2021), and Boerman et al. (2021), to ensure both the theoretical alignment and empirical robustness of the collected data. Data collection was conducted through Google Forms using a structured questionnaire distributed through various channels such as electronic mailing and professional networks. This methodology ensured the inclusion of participants with relevant knowledge of the website's cookie. Participation was voluntary, the questionnaire was designed to require approximately 10–15 min for completion, and responses were maintained in an anonymous format to uphold participant privacy.

A non-probability purposive sampling technique was used to collect data from participants. The sample size of 384 participants was determined using the Raosoft sample size calculator based on a 95 % confidence level and a 5 % margin of error. The sample size ensured the statistical validity of the findings for the population of interest. Participants were selected on the basis of their awareness of the website's

Table 18
Summary model for the moderating effect hypothesis (PSUSP).

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.464[a]	.216	.210	1.11634	.216	41.233	1	150	.000
2	.504[b]	.254	.244	1.09210	.039	7.732	1	149	.006

[a] Predictors: (Constant), PEOU.
[b] Predictors: (Constant), PEOU, PSUSP.

consent. This targeted sampling technique facilitated the collection of relevant data from knowledgeable respondents, thereby increasing the quality and depth of insights into attitudes toward website cookie consent in Jordan.

4.2. Questionnaire design and measures and scale development

The questionnaire was developed by adopting validated scales from previous research, to ensure that the measures were relevant to the constructs. The scales used to measure participants' attitudes and behaviors towards website cookie consent are illustrated in Tables 2–8 as follows:

Table 2 illustrates the items used to measure the participants' perceptions of *Perceived Usefulness* of website cookie consent. This study employed questions and metrics derived from Davis (1989). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 3 illustrates the items used to measure participants' perceptions of *Perceived ease of use* of website cookie consent. This study employed questions and metrics derived from Davis (1989). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 4 illustrates the items used to measure the participants' perceptions of *Perceived Severity*. This study employed questions and metrics adopted from Boerman et al. (2021). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 5 illustrates the items used to measure the participants' perceptions of *Perceived Susceptibility*. This study employed questions and metrics adopted from Boerman et al. (2021). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 6 illustrates the items used to measure the participants' perceptions of *Self-efficacy*. This study employed questions and metrics adopted from Boerman et al. (2021). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 7 illustrates the items used to measure the participants' perceptions of *Response Efficacy*. This study employed questions and metrics derived from Strycharz et al. (2021) and Tsai et al. (2016). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 8 illustrates the items used to measure participants' perceptions of *Cookie opt-in motivation.* This study employed items and metrics adopted from Strycharz et al. (2021) and Tsai et al. (2016). Participants were asked to indicate their level of agreement with the following statements on a 7-point scale, where 1 represented strong disagreement and 7 represented strong agreement.

Table 21
ANOVA for the moderating effect hypothesis (SE).

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	51.385	1	51.385	41.233	.000[b]
	Residual	186.931	150	1.246		
	Total	238.316	151			
2	Regression	56.896	2	28.448	23.365	.000[c]
	Residual	181.419	149	1.218		
	Total	238.316	151			

a. Dependent Variable: OPTIN.
[b] Predictors: (Constant), PEOU.
[c] Predictors: (Constant), PEOU, SE.

5. Data analysis and results

The collected data were analyzed using the Statistical Package for the Social Sciences (SPSS). Cronbach's alpha was calculated for each construct to examine the reliability of the scale. To verify the validity of the questionnaire, scholars and experts in the field conducted a face validity assessment. Regression analysis was used to test the relationships between the independent and dependent variables to accept or reject the hypotheses.

5.1. Demographic analysis

A summary of the sample profiles is presented as follows:

- As shown in Table 9, the age group of 21–30 years dominated the sample, comprising 46.1 % of the respondents, followed by 30.2 % those aged 20 years or less. The 31–40 years age group represented 15.9 %, with the 41–50 years and 51 years or above categories accounting for 6.5 % and 1.3 % of the population, respectively.
- Table 10 presents the respondents' educational levels. The majority of respondents held a bachelor's degree, accounting for 68.5 % of the total. Postgraduate degree holders made up 23.7 %, while those with a Diploma and Secondary school or fewer qualifications made up 3.9 %. This distribution indicates a high level of educational attainment among the respondents, particularly in higher education.

5.2. Descriptive analysis

A descriptive analysis of the data provided a comprehensive overview of the responses to a range of variables. Table 11 shows the participants' responses across independent variables (Perceived Usefulness and Perceived Ease of Use), moderating variables (Perceived Severity, Perceived Susceptibility, Self-efficacy, and Response Efficacy), and the dependent variable (Cookie Opt-In Motivation). Key statistics, including the sample size (N), minimum, maximum, mean, and standard deviation, provide valuable insights into the central tendencies and variations within the dataset.

Table 12 presents a descriptive statistical overview of the pivotal variables. The analysis of the constructs related to website cookie consent revealed varying perceptions among participants. For Perceived Usefulness (PU), the average rating was $\mu = 4.7237$ with a standard deviation of $\sigma = 1.11401$, indicating a generally positive perception of website usefulness, with moderate variability among responses.

Table 20
Summary model for the moderating effect hypothesis (SE).

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.464[a]	.216	.210	1.11634	.216	41.233	1	150	.000
2	.489[b]	.239	.229	1.10344	.023	4.527	1	149	.035

[a] Predictors: (Constant), PEOU.
[b] Predictors: (Constant), PEOU, SE.

Table 22
Summary model for the moderating effect hypothesis (RE).

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.464[a]	.216	.210	1.11634	.216	41.233	1	150	.000
2	.508[b]	.258	.248	1.08914	.043	8.585	1	149	.004

[a] Predictors: (Constant), PEOU.
[b] Predictors: (Constant), PEOU, RE.

Table 23
Coefficients for the moderating effect hypothesis (RE).

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.981	.412		4.810	.000
	PEOU	.577	.090	.464	6.421	.000
2	(Constant)	1.187	.485		2.449	.015
	PEOU	.527	.089	.424	5.895	.000
	RE	.230	.079	.211	2.930	.004

a. Dependent Variable: OPTIN.

Perceived Ease of Use (PEOU) was followed by a mean rating of $\mu = 4.4704$ ($\sigma = 1.01068$), showing a moderate level of PEOU perceived by participants with relatively consistent attitudes across responses. The perceived Severity (PS) was calculated as $\mu = 4.5263$ ($\sigma = 1.38239$).

reflecting a moderate perception of severity with noticeable variability in the views. Perceived Susceptibility (PSUSP) was rated higher with a mean of $\mu = 5.4605$ ($\sigma = 1.04634$), indicating a lower level of perceived threat from websites, with participants showing relatively consistent perceptions. Self-efficacy (SE) in managing cookie consent was moderate at $\mu = 3.8191$ ($\sigma = 1.19019$), showing variability in individuals' confidence levels. Response Efficacy (RE) had a mean score of $\mu = 4.5614$ ($\sigma = 1.25628$), suggesting a moderate belief in the effectiveness of managing cookie consent with some variation among participants. Finally, Cookie Opt-In Motivation (OPTIN) also had a mean score of $\mu = 4.5614$ ($\sigma = 1.25628$), indicating a moderate level of motivation towards accepting cookies with moderate variability in motivation levels among the participants. These findings collectively highlight a spectrum of attitudes towards website cookie consent, ranging from moderate to positive, with variances in individual perceptions and attitudes across constructs.

5.3. Reliability and validity

The reliability of the scales was evaluated based on Cronbach's alpha, using the scale reliability test in SPSS. According to Taber (2018), reliability is satisfied if the value of Cronbach's alpha is greater than 0.7. Table 13 shows that the value of Cronbach's alpha was .735 for the four items of PU, .884 for the five items of PEOU, .848 for the three items of PS, .739 for the three items of PSUSP,.859 for the four items of SE, .850 for the three items of RE, and .915 for the three items of OPTIN.

To verify the validity of the questionnaire, scholars and experts in the field conducted a face validity assessment. Face validity refers by Musarat et al. (2024) to how an instrument is able to measure the concept it was designed to measure by Musarat et al. (2024). If an instrument appears to measure the concept proposed for the measurement, it is supposed to have face validity. This aspect of validity can be constructed, significantly improved, and discussed with several modifications. For this study, most of the feedback focused on removing statements that lacked clarity, used informal language, did not specify the type of data collection (e.g., not clearly through website cookies), and contained ambiguous wording. All questions were refined according to the scholar's feedback, and the refined ones (mentioned in Tables 2–8) were used in the data collection.

5.4. Results

5.4.1. Direct effect hypotheses – H1 and H2

Table 14 shows that the correlation coefficient (R) of .484 indicates a moderate positive relationship between the independent variables — Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) — and the dependent variable, OPTIN. PU and PEOU account for 23.4 % of the variance in OPTIN ($R^2 = .234$). In Table 15, although PU showed a positive relationship with OPTIN ($B = .172$, $p = .060$), it was only marginally significant. In contrast, PEOU had a stronger and statistically significant influence on OPTIN ($B = .490$, $p < .001$), with a standardized coefficient of .394. This indicates that PEOU has a stronger effect over PU in influencing cookie consent behavior.

5.4.2. Moderating effect hypothesis

5.4.2.1. Perceived threat severity (PS) – H3. Tables 16 and 17 present the results of the regression analysis assessing the moderating effect of Perceived Threat Severity (PS) on the relationship between Perceived Ease of Use (PEOU) and Cookie Opt-In Motivation (OPTIN). In Table 16 Model 1, which includes PEOU as the sole predictor, explains 21.6 % of the variance in OPTIN ($R^2 = .216$) with an Adjusted R^2 of .210 and a standard error of 1.11634, indicating that ease of use significantly influences cookie opt-in behavior. In Table 17, PEOU has a significantly strong positive effect on OPTIN (unstandardized coefficient = .577, standardized coefficient = .464, both $p < .001$), demonstrating that higher perceived ease of use strongly increases the likelihood of opting in. When PS was added as a moderating variable in Model 2 (see Table 16), the explanatory power of the model improved, with R^2 increasing from .216 to .312. This means that the inclusion of PS accounts for an additional 9.7 % of the variance in OPTIN, reflecting its substantial role in the decision-making process. The R^2 change was statistically significant (F change = 20.939, $p < .001$), confirming that PS significantly moderated the effect of PEOU on OPTIN. The negative coefficient for PS (unstandardized coefficient = − .290, standardized coefficient = − .319, both $p < .001$, as per Table 17) suggests that as users perceive higher levels of threat severity, the positive impact of PEOU on opting diminishes. This indicates that users are less likely to opt-in to cookies when they feel threatened by potential privacy risks,

Table 24
Summary of the discussion on H1 and H2.

Hypothesis	Description	Result	Empirical Finding
H1: Perceived Usefulness (PU) Influence	The belief that tracking cookies enhance user experience will motivate opt-in decisions.	Rejected	Weaker than expected, suggesting the need for clearer benefits communication.
H2: Perceived Ease of Use (PEOU) Influence	The ease of managing cookie preferences will positively affect opt-in decisions.	Accepted	Strong influence, indicating that simplification of preference management is crucial.

even if the consent mechanism is easy to use.

5.4.2.2. Perceived susceptibility (PSUSP) – H4. Tables 18 and 19 present the results of the regression analysis assessing the moderating effect of Perceived Susceptibility (PSUSP) on the relationship between Perceived Ease of Use (PEOU) and Cookie Opt-In Motivation (OPTIN). As shown in Table 18, Model 1, which includes PEOU as the sole predictor, explains 21.6 % of the variance in OPTIN ($R^2 = .216$). According to Table 19, the unstandardized coefficient for PEOU is B = .577, while the standardized coefficient is B = .464, $p < .001$, indicating a strong positive effect on opt-in behavior. In Table 18 Model 2, after introducing PSUSP as a moderating variable, the explained variance increased to 25.4 % ($R^2 = .254$), with an Adjusted R^2 of .244. An R^2 change of .039 and a significant F change of 7.732 ($p = .006$) confirmed the significant moderating role of PSUSP. The unstandardized coefficient for PSUSP is B = .238, while the standardized coefficient is B = .198, p = .006 (as per Table 19), indicating that users with higher perceived susceptibility to privacy risks are more likely to opt-in, especially when they find that the consent mechanism is easy to use.

5.4.2.3. Self-efficacy (SE) – H5. The regression analysis presented in Table 20 examines the moderating effect of Self-Efficacy (SE) on the relationship between Perceived Ease of Use (PEOU) and Cookie Opt-In Motivation (OPTIN). The results were reported using two models. In Model 1, with PEOU as the sole predictor, the model showed a moderate positive correlation between PEOU and OPTIN (R = .464). This model explains 21.6 % of the variance in OPTIN ($R^2 = .216$) with an Adjusted R^2 of .210, indicating that PEOU is a significant predictor of opt-in motivation. The significance of the model was confirmed by an F-value of 41.233 ($p < .000$), as shown in the ANOVA results in Table 21, establishing that PEOU had a statistically significant impact on OPTIN.

In Model 2, after introducing SE as a moderating variable, the explanatory power of the model improved slightly (see Table 20). The R value increases to.489, and the model explained 23.9 % of the variance in OPTIN ($R^2 = .239$). The Adjusted R^2 increases to.229, reflecting this improvement after adjusting for the number of predictors. The inclusion of SE resulted in an R^2 change of .023 and an F change of 4.527 ($p = .035$), indicating a small but statistically significant contribution of SE as a moderator. This finding suggests that SE has a modest impact on the relationship between PEOU and OPTIN. The ANOVA results in Table 21 further confirm the statistical significance of Model 2, with an F-value of 23.365 ($p < .000$), indicating that the combined effect of PEOU and SE on OPTIN was significant. This analysis indicates that while PEOU remains the primary predictor of opt-in behavior, the inclusion of SE as a moderator slightly increases the explanatory power of the model, highlighting that users' confidence in managing cookie settings (i.e., SE) slightly enhances their likelihood of choosing, although the effect is less substantial than that of other factors.

Table 25
Summary of the discussion on H3 to H6.

Hypothesis	Description	Result	Empirical Finding
H3: Perceived Threat Severity (PS) Moderation	Higher Perceived Threat Severity reduces the positive impact of PEOU on OPTIN.	Accepted	Design consent mechanisms that effectively communicate the security measures in place to protect user data, reducing perceived threats.
H4: Perceived Susceptibility (PSUSP) Moderation	PSUSP alongside PEOU enhances OPTIN decisions.	Accepted	Incorporate elements that address users' susceptibility concerns, such as detailed privacy information and customization options.
H5: Self-Efficacy (SE) Moderation	SE slightly but significantly moderates the PEOU-OPTIN relationship.	Accepted	Develop user interfaces that are not only easy to use but also empower users, making them feel competent in managing their privacy settings.
H6: Response Efficacy (RE) Moderation	RE positively influences the PEOU-OPTIN relationship.	Accepted	Ensure that the consent interface clearly outlines how opting in or out will specifically impact the user's privacy and data security.

5.4.2.4. Response efficacy (RE) – H6. Tables 22 and 23 present the results of the regression analysis assessing the moderating effect of Response Efficacy (RE) on the relationship between Perceived Ease of Use (PEOU) and Cookie Opt-In Motivation (OPTIN) is presented across two models. As reported in Table 22, Model 1, which includes PEOU as the sole predictor, shows a moderate and significant correlation with OPTIN (R = .464), explaining 21.6 % of the variance in OPTIN ($R^2 = .216$, Adjusted $R^2 = .210$). In Model 2, after introducing RE as a moderating variable, the model's explanatory power improved, with R increasing to.508, and the model explained 25.8 % of the variance in OPTIN ($R^2 = .258$). The Adjusted $R^2 = .248$, R^2 change of .043, and F change of 8.585 ($p = .004$) confirmed the significant contribution as a moderating variable.

The coefficients in Table 23 further show that PEOU maintains a strong positive effect on OPTIN (unstandardized coefficient = .527, standardized coefficient = .424, $p < .000$), and the introduction of RE (unstandardized coefficient = .230, standardized coefficient = .211, p = .004) suggests a substantial additional influence. The positive coefficient for RE suggests that higher levels of response efficacy amplify the impact of ease of use on opt-in behavior. These findings emphasize that while PEOU is important, users' beliefs about the effectiveness of their actions (i.e., RE) significantly shape their willingness to opt-in.

6. Discussion

Although TAM and PMT have been extensively used to study user behavior in technology adoption and privacy settings, there is a notable absence of research combining these frameworks with DSR. This integration is crucial for developing user-centered consent mechanisms that are both effective and compliant with regulatory standards, such as the GDPR. The findings of this study offer insightful contributions to the understanding of user behavior regarding opt-in decisions for tracking cookies, particularly within the scope of Information Systems research, as follows:

6.1. Direct hypotheses (H1 and H2)

In the Technology Acceptance Model (TAM), Perceived Usefulness (PU) reflects users' beliefs that a technology, such as tracking cookies, will provide value through benefits such as personalized content or enhanced website functionality. However, users often fail to see these benefits owing to transparency issues, complex privacy policies, and distrust towards data usage (Kulyk et al., 2018), leading to a lower preference for opting in. Hypothesis H1 posits a direct relationship between PU and motivation to accept cookies, but empirical findings suggest that this relationship is weaker than expected due to the aforementioned factors.

Conversely, Perceived Ease of Use (PEOU) in TAM refers to how effortlessly a technology is to use. To track cookies, PEOU relates to users' ability to understand and control their cookie preferences. We posit that users often find cookies difficult and opt for the easiest option, which typically involves accepting them. The findings of this study revealed a stronger influence of PEOU on PU in cookie consent decisions, supported by the significant coefficients in the model analysis.

This behavior is in line with the findings of Giese and Stabauer (2022) where users prefer to quickly dismiss cookie notices, reflecting a tendency to prioritize convenience over informed decision-making. Table 24 summarizes the main findings for H1 and H2.

The focus on cookie consent in Jordan provides insights into how cultural factors affect behavior. In collectivist cultures such as Jordan, where trust in institutions and social harmony are valued, users prioritize ease and convenience over privacy concerns when consenting to cookies (Hofstede, 2001). Similar patterns have been observed in other collectivist cultures such as Pakistan and Turkey (Abbasi et al., 2015). This contrasts with the individualistic cultures in Western Europe and North America (Hofstede, 2001), where users prioritize privacy, often reject non-essential cookies, and seek value derived from data tracking and tailored content (Boerman et al., 2021; Degeling et al., 2019). Although this study targeted participants in Jordan with pre-existing awareness of cookie policies, PEOU emerged as a significant predictor of cookie consent acceptance, indicating that even digitally aware users in Jordan favor ease and simplicity over personalization when consenting to cookies.

6.2. Moderating effect hypotheses (H3–H6)

Regression analyses for hypotheses H3–H6 provided valuable insights into the moderating effects of psychological factors on the relationship between PEOU and users' decisions to opt-in (OPTIN).

For H3, perceived threat severity (PS) plays a significant role in diminishing the positive influence of PEOU on opt-in behavior. While ease of use alone explains a moderate portion of the variance in opt-in decisions, a higher PS reduces users' willingness to opt-in. This indicates that users become more risk averse as their perception of privacy threats increases, leading them to prioritize safety over convenience, even when the consent process is simple. In contrast, Perceived Susceptibility (PSUSP) enhances the impact of PEOU on opt-in motivation (H4). Users who feel more vulnerable to privacy risks are more inclined to opt for a process that is easier to navigate. This suggests that users with higher perceived susceptibility are more likely to take protective actions, such as accepting cookies, if they believe it mitigates potential threats, and if the process does not require significant effort.

These findings have added significance in the cultural context of Jordan, where trust in institutions and preferences for convenience are predominant. A higher PS erodes institutional trust, making perceived risks too great to ignore, even in a society that traditionally favors institutional authority. On the other hand, the PSUSP aligns well with Jordanian cultural values that prioritize simplicity and ease. When the consent mechanism is easy to use, users feel empowered to manage privacy risks with minimal effort, reinforcing their preference for convenience and practicality in decision-making.

In Hypothesis H5, Self-Efficacy (SE) also moderates the relationship between PEOU and opt-in behavior, although with a more modest effect. This indicates that although ease of use remains a fundamental predictor, the findings suggest that users with greater confidence in their ability to manage cookie settings are more likely to engage in the consent process. Finally, Hypothesis H6 revealed the significance of Response Efficacy (RE) as a moderating variable. The findings revealed that RE strengthens the relationship between PEOU and opt-in motivation, as users are more likely to opt in when they believe that their actions—whether accepting or rejecting cookies—will effectively protect their privacy. This emphasizes that beyond ease of use, users' belief in the effectiveness of their actions enhances their willingness to opt in. This is particularly relevant in contexts where the perceived benefits of using technology play a crucial role in decision making, such as health informatics, cybersecurity, and e-commerce.

When comparing these findings to those from other regions, cultural, legal, and educational differences are important factors to consider. For instance, in Jordan, users have reported moderate levels of SE and RE when managing cookie settings. In contrast, studies conducted in Northern Europe and the United States, where privacy education is more extensive, show higher SE and RE levels. Users in these regions tend to be more empowered to navigate privacy settings effectively, which often results in more selective consent behaviors (Tsai et al., 2016; Zhang et al., 2020). These cultural differences underscore the need for future research to account for regional variations when generalizing results beyond Jordan. Effective and user-friendly cookie consent mechanisms must be designed with these cultural and educational contexts in mind to suit diverse populations better. Table 25 summarizes the main findings for H3–H6.

6.3. Requirements-focused DSR

This section integrates the findings of the study into the DSR framework, emphasizing the value of design-centric approaches that address user concerns and the requirements for cookie consent. This approach contributes to the development of transparent and user-friendly cookie consent mechanisms. This study centers on Requirements-Focused DSR, which aims to design cookie consent mechanisms that enhance users' perceptions, thereby increasing the likelihood of informed opt-in decisions.

6.3.1. Design requirements for PU and PEOU

Enhance transparency to communicate benefits (DR1)

Empirical result: The findings indicate that PU has a lower impact on opt-in motivation compared to PEOU, primarily due to a lack of transparency in cookie consent forms.

Practical recommendations:

Practitioners should employ clear and concise language to explain how cookies enhance their browsing experience by improving personalization and performance. For instance, statements such as "Cookies help us remember your preferences and enhance content", "We use cookies to enhance your experience. Click here to customize your preferences or accept", or "We use cookies for functionality and personalization. Clicking here to learn more" can explicitly communicate these benefits to users while reducing complexity (Kulyk et al., 2018).

- Simplify the interface for enhanced usability (DR2)

Empirical result: PEOU is a significant predictor of cookie acceptance. Thus, simplifying the cookie consent interface is essential for enhancing user engagement and promoting informed decision making.

Practical recommendations:

To enhance usability, cookie consent interfaces should adopt a clean and minimalist design to avoid complex text layout. The consent banner should be positioned as a slim, fixed bar at the bottom of the screen to minimize intrusiveness, while maintaining visibility. Additionally, displaying the banner only after initial user engagement, such as scrolling or clicking, allows users to become familiar with the website before encountering consent options, thereby reducing their cognitive load (Giese and Stabauer, 2022).

Furthermore, the consent color scheme should employ neutral tones that match the website's design, whereas action buttons should use contrasting colors to make the consent noticeable, without interrupting the user experience. For optimal functionality, practitioners should ensure that the cookie consent interface is adaptable to smaller screens so that it remains compact, swipe-able, and does not obstruct the site's core content. To further simplify user choices, the consent should present minimal yet clear options, such as "I allow," "No, thank you," and "Customize" replacing technical terminology with friendly, approachable language. Visual cues can also enhance clarity, using a thumbs-up icon for "I allow," a thumbs-down icon for "No, thank you," and a gear icon for "Customize" can ensure an intuitive and user-friendly interaction.

6.3.2. Design requirements addressing moderating variables (PS, PSUSP, SE, RE)

- Address perceived threat severity with clear security information (DR3)

Empirical result: PS reduces the positive influence of PEOU on opt-in motivation. Users who perceive higher threats to their privacy tend to hesitate in consenting to cookies, unless they receive explicit reassurance regarding data security.

6.3.2.1. Practical recommendations. Practitioners should incorporate recognized trust symbols, such as "GDPR Compliant" badges, within consent banners to signify adherence to data protection standards, thereby reinforcing trust without occupying excessive space. Additionally, brief security messages such as "Your data is secured with industry-standard encryption" can reassure users by emphasizing data protection measures. For those seeking more information, a clearly labeled "Learn More" or "How we protect your data" link should provide direct access to a detailed page explaining privacy measures such as encryption, anonymization, and data minimization practices. Furthermore, utilizing consistent, reassuring language throughout the consent banner (e.g., "your data is safe with us" or "we prioritize your privacy") can further reduce perceived threat severity and help foster a secure and transparent user experience.

- Acknowledge user susceptibility and enhance privacy control (DR4)

Empirical result: PSUSP positively moderates the effect of PEOU on opt-in when users feel empowered to manage privacy risks. Interfaces must allow users to easily manage their preferences and understand how their choices affect privacy and data security.

6.3.2.2. Practical recommendations. Consent interfaces should employ a progressive disclosure design to balance the ease of use with detailed control. Presenting initial options, such as "Accept All" or "Customize," provides users with straightforward choices, while additional, detailed settings accommodate those seeking greater privacy control. This approach helps reduce cognitive load while still respecting the needs of privately conscious individuals.

For users who choose to customize, granular options through toggle switches or checkboxes for categories such as essential, performance, analytics, and advertising cookies are recommended. Instead of using a binary accept/reject choice, this granular approach allows users to select specific types of cookies while rejecting others. Each category should include concise, plain language explanations to help users understand the implications of their choice. For example, "Essential cookies keep the site running securely; they do not track personal data", "Performance cookies enhance site speed, using data that does not identify you," "Analytics cookies improve your experience without revealing who you are," and "Advertising cookies show you ads that matter to you and are designed not to reveal your identity." Alternatively, if practitioners believe that these explanations compromise consent's user-friendliness, they can display these messages through a tooltip accessible by tapping an icon (e.g., an "i" or question mark symbol) next to each cookie category. Tapping on the icon opens the tooltip, and tapping outside the tooltip closes it. Moreover, to support users' ongoing control over their privacy settings, a small, persistent "Cookie Settings" link should be included in prominent and easily accessible areas, such as the website's footer or within user profile sections. This inclusion minimizes the need for repeated consent banners and provides users with convenient access to their preferences, allowing them to revisit and modify settings as desired without disrupting their browsing experience.

- Increase self-efficacy through user education and accessible options (DR5)

Empirical result: SE has a moderate but positive influence on opt-in behavior. Users are more willing to consent if they feel capable of managing their privacy settings and confident in their ability to protect their data.

6.3.2.3. Practical recommendations. To enhance self-efficacy without compromising ease of use, the initial consent banner should include a link directing users to a dedicated privacy-setting page (as mentioned earlier in DR3). This page can house interactive elements such as FAQs, step-by-step guides, and tutorials that visually educate users on managing their privacy settings and understanding data usage. Completing these resources could reward users with a "Privacy Champion" badge, fostering confidence through engaging and educational experience (i.e., using game-design elements in non-gaming contexts). Providing these resources encourages adoption among users who might be hesitant to engage with new technologies owing to their limited technical skills. Additionally, hosting these resources on a separate page allows users to access in-depth support as needed, keeping the primary consent interface streamlined and user friendly. Reinforcing messages such as "Control your data" instead of "Customize" can further empower users by emphasizing their control over their privacy.

- Communicate response efficacy to reinforce trust (DR6)

Empirical result: The findings revealed that RE strengthens the motivation to opt in because users are more willing to consent when they feel that their actions have a tangible impact on their privacy.

6.3.2.4. Practical recommendations. In addition to displaying a concise explanation of each cookie category, as discussed in Section DR4, consent interfaces should provide immediate real-time feedback when users adjust their cookie settings. For instance, displaying a confirmation message such as "Your privacy settings have been saved" or "Your preferences have been updated to allow only essential cookies" reassures users that their actions have been successfully implemented. Additionally, integrating a visual indicator that reflects a user's current privacy status can further reinforce the impact of their choices. For example, an animated dynamic icon (e.g., a lock closing, shield filling up, or progress bar) could be changed based on the chosen cookie settings. As users allow or reject certain cookie types, this dynamic icon updates to show their "Privacy Level" with labels like "Maximum Privacy" for essential-only settings and "Enhanced Experience" for all cookies. This visualization reinforces RE by making the impact of consent choices visible, thus providing users with a quantifiable sense of control over their privacy.

6.4. Theoretical implications

This study presents the first interdisciplinary theoretical approach by integrating the TAM to examine technology acceptance and adoption, PMT to examine the choices regarding a protective action in response to perceived threats or security risks, and DSR to follow a systematic approach to propose a set of requirements in cookie consent research. While most of the previous studies primarily investigated choice architecture manipulation, dark pattern design, and nudging – refer to Table 1 for a summary of previous studies–there remains a research gap on balancing privacy and usability in cookie consent research. Therefore, our study proposes a user-centric approach to understand privacy and usability decision-making to propose GDPR-compliant cookie consent mechanisms that balance regulatory and user-centric requirements. The findings of this study contribute to the theoretical knowledge as follows.

- The role of PMT theory in GDPR compliance for cookie consent mechanisms is important because it influences user behavior based on GDPR compliance, including coping and threat appraisal factors. In addition to user rights such as data protection, the right to have their data forgotten, transparency, and data portability.
- The role of TAM in cookie consent mechanisms is important by influencing users' decision to opt-in or opt-out.
- The role of the requirement-focused design approach is to produce ethical, transparent, and user-friendly cookie consent mechanisms that comply with GDPR requirements and allow users to make informed decisions while browsing websites.
- The role of interdisciplinary theoretical integration between TAM, PMT, and DSR. This interdisciplinary perspective has proven that cookie consent mechanisms should not only rely on regulatory compliance, but also on mechanisms that address usability and psychological factors.

Researchers can use the proposed interdisciplinary theoretical approach in this study and apply it in various technological contexts such as: AI-related consent mechanisms, blockchain data sharing agreements, mobile applications, and topic API for interest-based advertising.

6.5. Open innovation in the design of website cookies consents

As a result of rapid technological development, organizations adopt open innovation to establish novel ideas from internal resources and integrate external resources (Chesbrough et al., 2014). Innovation that originates from outside the organization (e.g., user involvement) is referred to as inbound and outbound innovation from inside the organization. Open innovation was first triggered by Chesbrough (2003) and it is defined as "a paradigm that assumes that firms can and should use external ideas as well as internal ideas, and internal and external paths to market, as they look to advance their technology." The open innovation concept has received increasing attention from scholars because of many factors, such as the reduction of the product life cycle, length and cost development processes, and value co-creation (Rumanti et al., 2021). Open-source software (OSS) is an example of adopting open innovation in the software development industry, such as outsourcing, global software development, and crowdsourcing (Munir et al., 2016). By accessing external resources for knowledge transfer, the application of open innovation in OSS increases innovative solutions and product quality, and reduces the time required to introduce OSS products to the market (Munir et al., 2018; Trzeciak et al., 2022).

Rumanti et al. (2021) investigate the impact of open innovation on SMEs performance through a systematic literature review. The study reported that open innovation has been implemented in various industries and improved organizational performance in several technology-driven domains. Therefore, we posit that the design science research adopted in this study is an application of the open innovation concept as it involves collecting requirements and proposing design solutions based on user involvement. In the context of this research, the design requirements for website cookie consent by users reflect an open approach that includes diverse novel sources of requirements as an inbound open innovation. These website cookie consent users represent market and consumer perspectives, offer unique insights into design requirements beyond organizational boundaries, and together create and share knowledge, novel ideas, and technologies through an open innovation approach (Yang et al., 2024; Gajdzik et al., 2024).

The design requirements of website cookies as software solutions that require development can be indicators of openness and knowledge transfer. Website cookie consent mechanisms can be of particular interest to organizations within related industries. Effective and innovative approaches to developing website cookie designs, such as transparent consent interfaces, clear control options, and compliance with privacy regulations, such as GDPR, can become a body of knowledge as a best practice and de facto industry standard. These design solutions enhance user trust and regulatory compliance and serve as a reference for the future development of website cookie consents, data collection, and privacy compliance across industries.

7. Conclusion

This study examined factors influencing users' acceptance of cookie consent using the Technology Acceptance Model (TAM), Protection Motivation Theory (PMT), and Design Science Research (DSR). With growing concerns over online privacy and the introduction of regulations, such as the GDPR, it is crucial to understand users' decisions regarding cookie consent. Through an analysis of 384 individuals familiar with cookie policies, this study identified the significant roles of perceived usefulness, ease of use, and threat and coping appraisals in shaping users' attitudes and intentions towards cookie consent acceptance. This study found that perceived ease of use is more influential than perceived usefulness in encouraging users to accept cookies. This highlights the importance of designing cookie consent mechanisms that are user-friendly, transparent, and empowering; address users' privacy concerns; and enhance their control over personal data. Additionally, this study explored the moderating effects of perceived severity, susceptibility, self-efficacy, and response efficacy, offering fresh insights into the interaction of these factors with the core constructs of TAM in the context of cookie consent.

This study utilized Design Science Research (DSR) to promote innovative cookie consent designs that balance ease of use with comprehensive privacy protection. By integrating TAM, PMT, and DSR, this research not only addresses a significant gap in the literature but also lays the groundwork for future investigations into the convergence of technology acceptance, privacy concerns, and regulatory compliance. As digital landscapes continue to evolve and privacy concerns intensify, insights from this study will play a crucial role in shaping the future of online consent mechanisms, ensuring that they are both user-centric and compliant with global privacy standards.

The findings of this study extend beyond academia by offering valuable insights for practitioners and policymakers seeking to create more effective and privately preserved online environments. These findings suggest several policy directions to enhance cookie consent practices by focusing on transparency, user control, and trust. To improve transparency, policies could mandate consent interfaces to employ clear plain language to explain the benefits of cookies, such as improved personalization, thereby supporting informed consent. Simplified, mobile-friendly designs offering straightforward options such as "I allow," "No, thank you," and "Control your data" would improve usability by reducing decision-making barriers and enhancing accessibility to a diverse user base. Additionally, regulatory guidelines could require the use of trust symbols (e.g., "GDPR Compliant") and concise security messages (e.g., "Your data is secured") to address privacy concerns and build user trust.

To foster user control, policies might enforce granular consent options with progressive disclosure design, enabling users to select specific cookie types or delve into detailed settings, as desired. Furthermore, policies promoting user self-efficacy could support the inclusion of educational resources within consent interfaces, allowing users to better understand privacy settings. Gamified elements, such as "Privacy Champion" badges, could further engage users and encourage active privacy management. Real-time feedback features such as confirmation messages (e.g., "Your settings have been saved") and privacy level indicators can reinforce users' choices by visibly affirming the impact of their actions (i.e., response efficacy). Lastly, policies could advocate for contextual consent prompts and a persistent "Cookie Settings" link accessible on all pages, giving users the flexibility to adjust preferences at any time. Collectively, policy developments in these areas would enhance user empowerment, improve the clarity of consent, and align digital privacy practices with evolving user expectations.

This study has some limitations, particularly the use of a non-probability purposive sampling method. As a result, the findings can only be generalized to individuals within a specific population who share characteristics with the sample, rather than the broader Jordanian population (Andrade, 2021). While the purposive sample provided targeted insights that enriched the research, future studies could benefit from a broader and more diverse participant base to expand the generalizability of these findings. In addition, future research can employ advanced statistical analysis techniques such as PL-SEM to handle complex cause-and-effect relationships with latent variables, hierarchical regression to examine the incremental impact of the variables by adding predictors in blocks and testing their impact toward user behavior on cookie consents, and running behavioral data in machine learning models (e.g., random forests and support vector machines) to predict which users are likely to opt-in or opt-out website cookies.

Moreover, given the diversity in user age, digital literacy, education, and cultural background, further research could explore the intersection of user demographics and technology acceptance. Specifically, future studies should address the following questions. (1) How do demographic factors such as age, education, and digital literacy affect users' understanding of cookie consent information, particularly regarding perceived ease of use and perceived usefulness? (2) Do cultural and educational differences influence the degree to which users trust and engage in cookie consent mechanisms? (3) How does digital literacy moderate the relationship between perceived ease of use and consent rates across various demographic segments? To investigate these research questions, mixed method approaches that combine quantitative surveys with qualitative interviews or focus groups segmented by demographic variables would offer valuable insights. Longitudinal studies could provide additional depth by tracking changes in user behavior and perceptions over time with increased exposure to cookie consent practices. In addition, experimental designs, in which different consent interface designs are tested across diverse user groups, could further clarify how demographic factors influence technology acceptance and engagement in privacy settings. By examining these demographic dimensions, future research could enhance our understanding of how diverse user characteristics shape the acceptance and effectiveness of cookie consent mechanisms, ultimately informing the design of more accessible, user-centered privacy tools.

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Ethical statement

The authors declare that this article is original, has not been submitted elsewhere, and adheres to all the ethical guidelines. All potential conflicts of interest have been disclosed, and the data presented are accurate and free of fabrication or manipulation. This research complies with all ethical standards and appropriate citations have been provided. The authors take full responsibility for the content and ethical compliance of this study.

CRediT authorship contribution statement

Abdallah Ammar: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Said Belal:** Validation, Software, Project administration, Formal analysis. **Ahmad Ala'eddin:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

AI-assisted technologies were used for English language editing during the preparation of this article. The authors reviewed and edited the manuscript as needed and took full responsibility for the content of the article.

Declaration of Competing Interest

The authors certify that they have no conflicts of interest related to the work reported in this article. The author has no commercial or other relationship with any of the respondents involved in the study. All respondents involved in the study voluntarily supported the academic research and did not receive any compensation for any kind.

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We are committed to maintaining transparency and uphold the highest standards of ethics in publishing. If further information or clarification is needed, please feel free to contact us.

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